

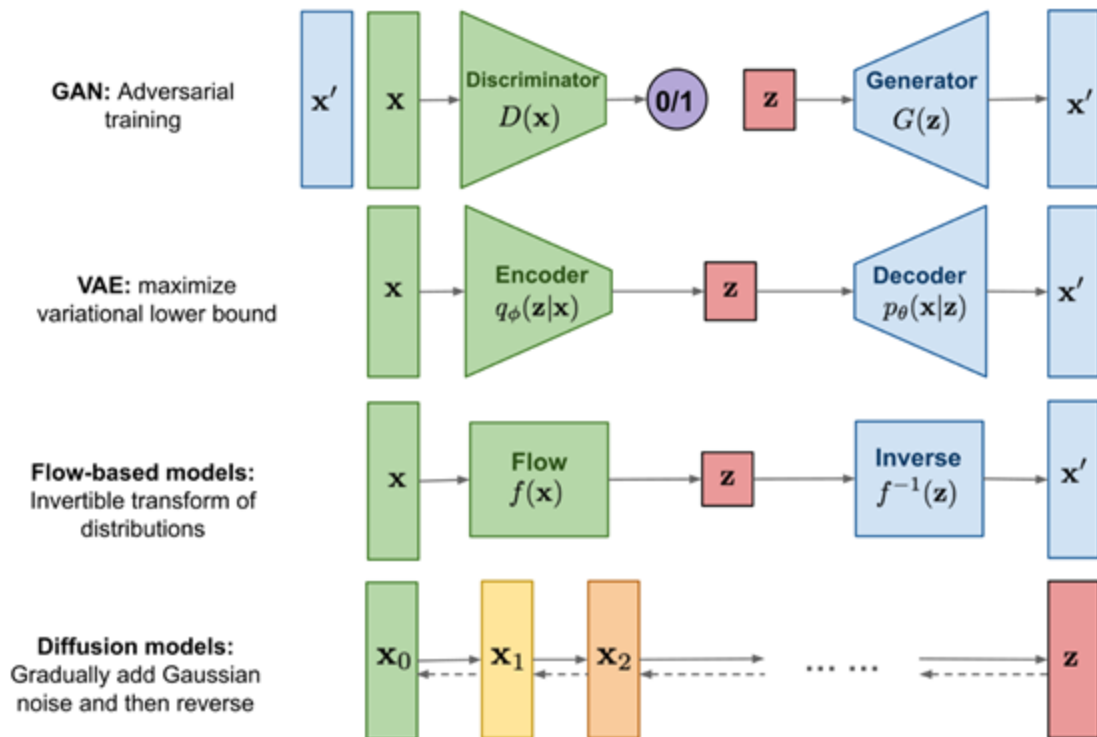
Diffusion Models for Audio Generation

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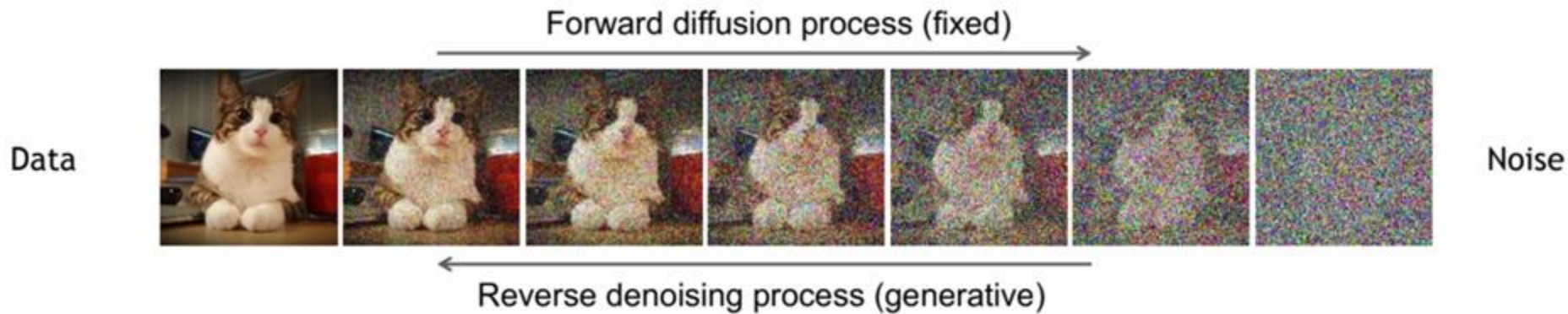
Agenda

- Introduction to Diffusion Models (a Multi-Perspective Approach):
 - Score Matching Perspective
 - Stochastic Differential Equations (SDE) Perspective
 - Latent Probabilistic Models Perspective
- Diffusion Models in CV:
 - DDPM, StableDiffusion, Diffusion Transformers (DiTs)
- Diffusion Models in Audio:
 - Vocoding: WaveGrad (2020), DiffWave (2020)
 - TTS: Diff-TTS (2021), GradTTS (2021)
 - Latent diffusion TTS: NaturalSpeech 2 (2023), NaturalSpeech 3 (2024), SimpleSpeech (2024)
 - Diffusion Transformer TTS: DiTTo-TTS (2024)

Generative models

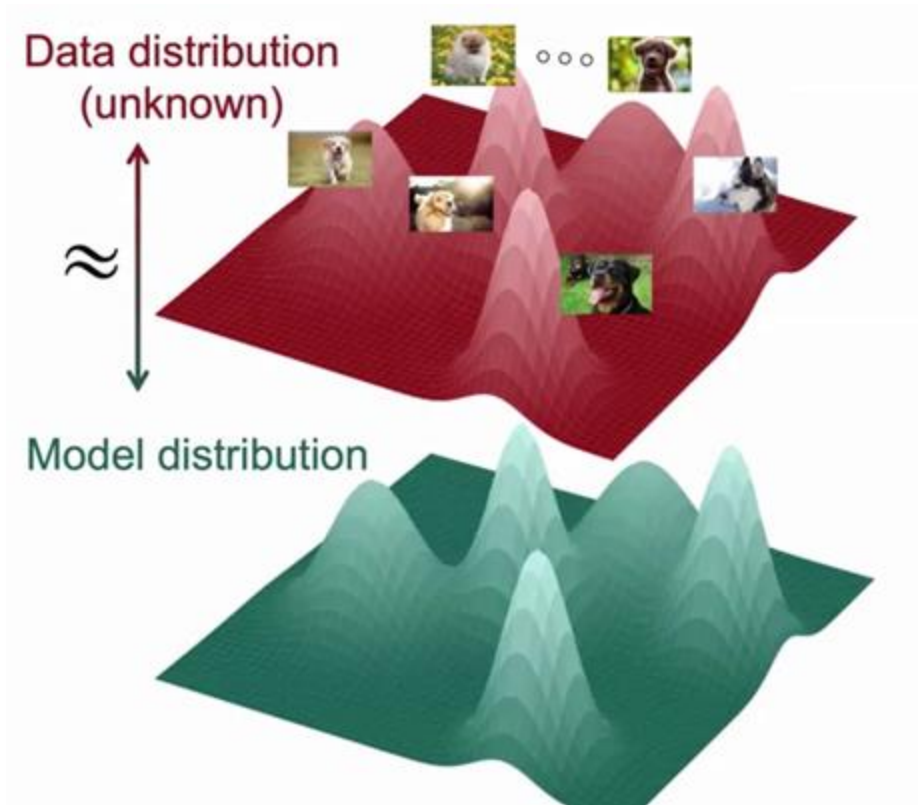


Diffusion models

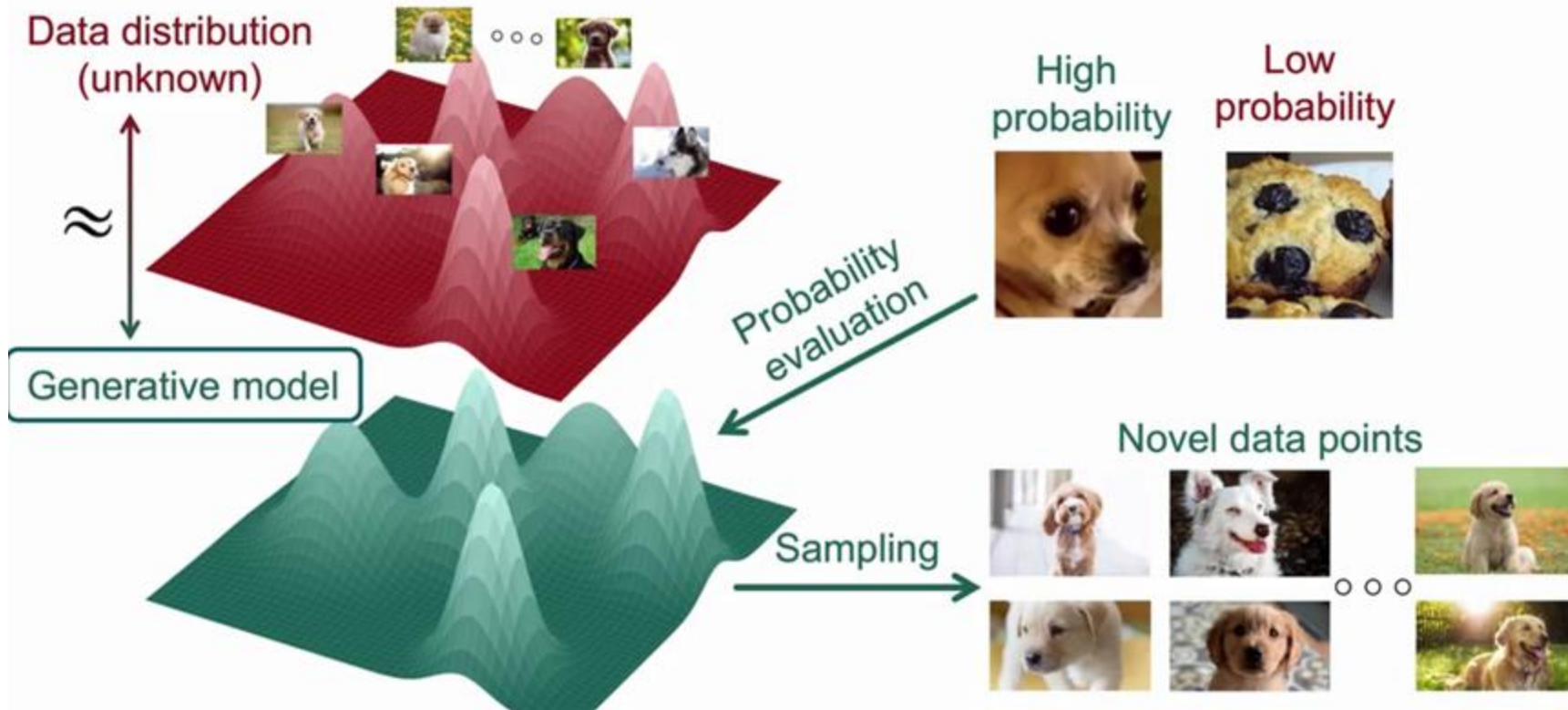


Score Matching Perspective

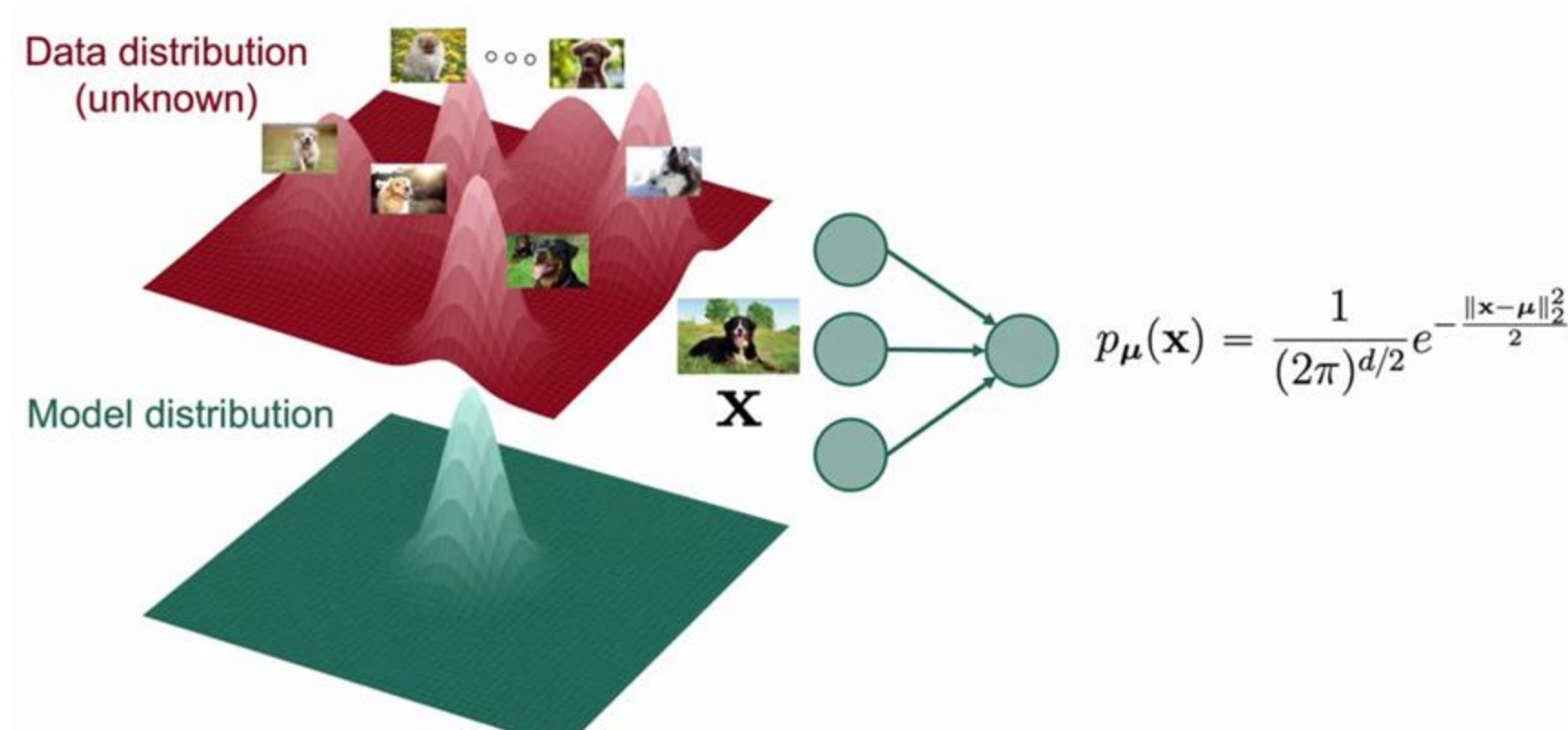
Estimating the probability distribution of data



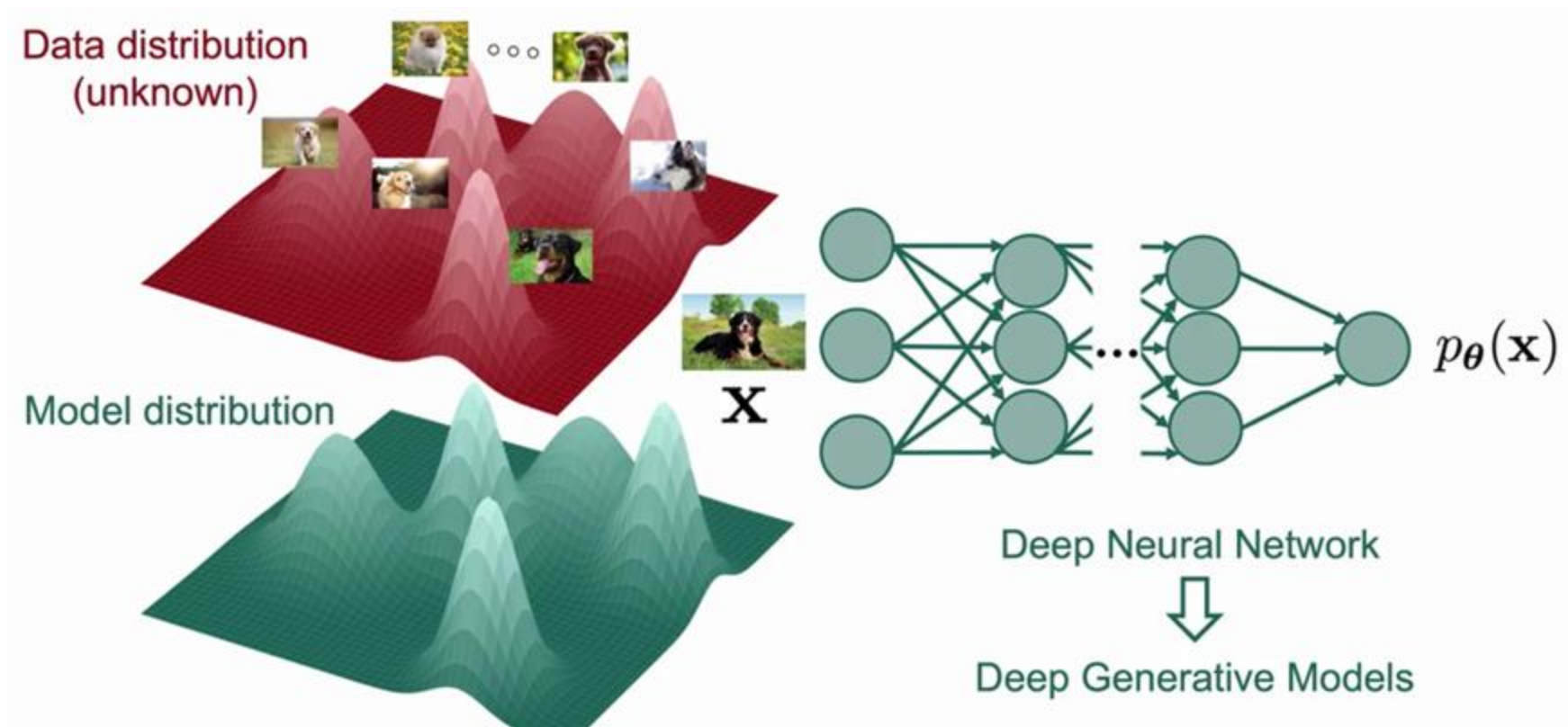
Estimating the probability distribution of data



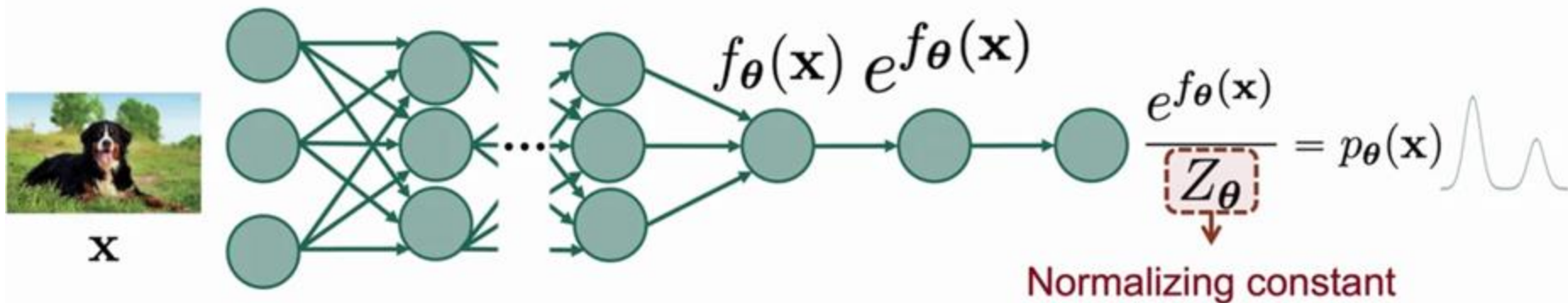
Estimating the probability distribution of data



Estimating the probability distribution of data



Estimating the probability distribution of data



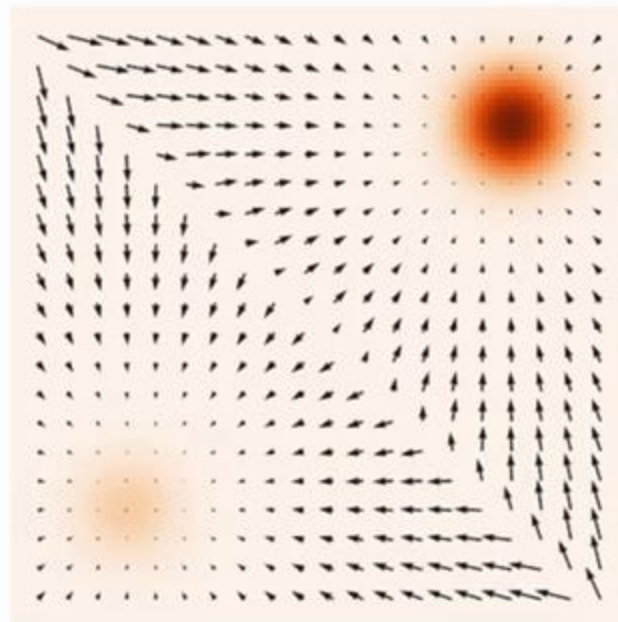
$$Z_{\theta} = \int e^{f_{\theta}(\mathbf{x})} d\mathbf{x}$$

Estimating the score function of data

$p(\mathbf{x})$
Probability density function



$\nabla_{\mathbf{x}} \log p(\mathbf{x})$
(Stein) score function



Score vs. density function

Estimating the score function of data



Training data

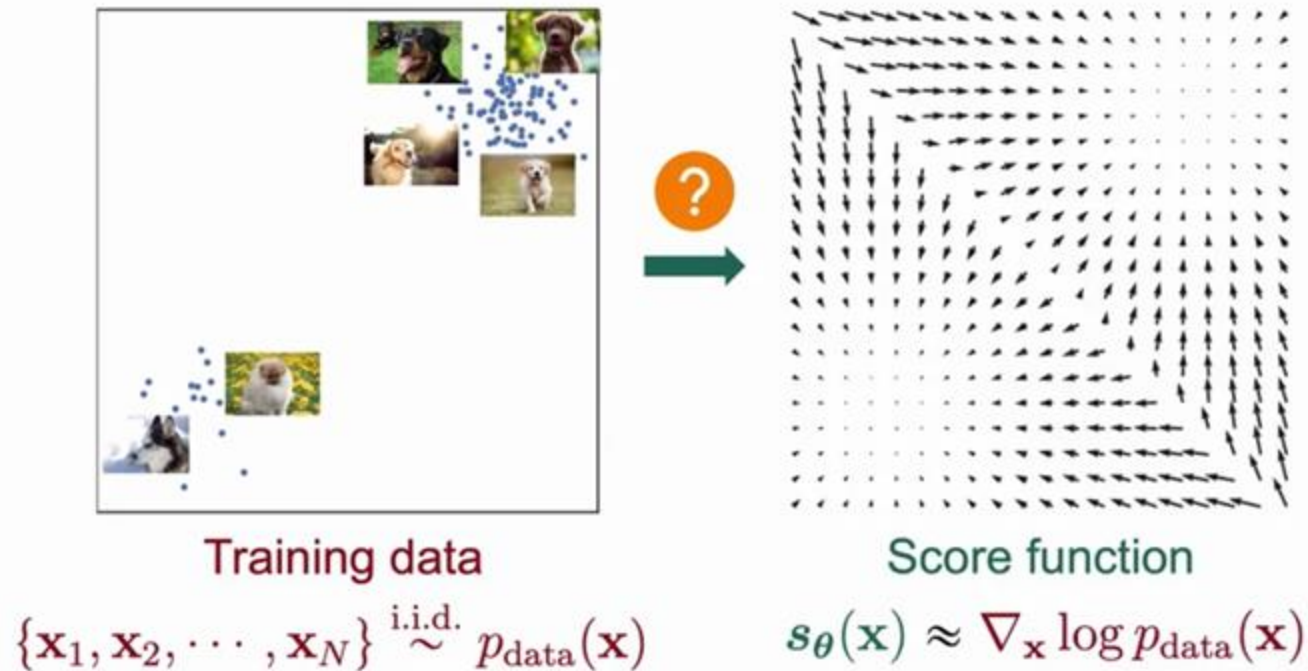
$$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}(\mathbf{x})$$



Probability density function

$$p_{\theta}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$$

Estimating the score function of data



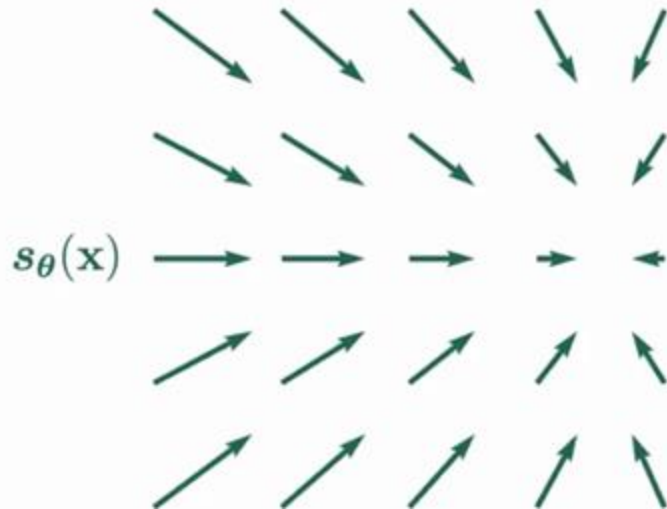
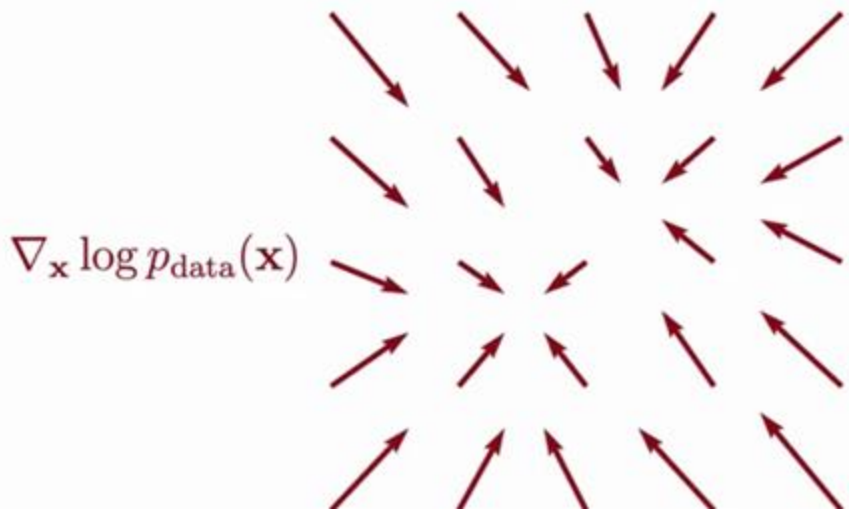
Estimating the score function of data

Given: $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}(\mathbf{x})$

Goal: $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Score Model: $s_{\theta}(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^d \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Objective: How to compare two vector fields of scores?



Estimating the score function of data

Given: $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}(\mathbf{x})$

Goal: $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Score Model: $\mathbf{s}_{\theta}(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^d \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Objective: How to compare two vector fields of scores?

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x})\|_2^2]$$

(Fisher divergence)

Estimating the score function of data

$$\begin{aligned} & \frac{1}{2} \mathbb{E}_{p_{\text{data}}} [(\nabla_x \log p_{\text{data}}(x) - \nabla_x \log p_{\theta}(x))^2] \\ &= \frac{1}{2} \int p_{\text{data}}(x) (\nabla_x \log p_{\text{data}}(x) - \nabla_x \log p_{\theta}(x))^2 dx \\ &= \underbrace{\frac{1}{2} \int p_{\text{data}}(x) (\nabla_x \log p_{\text{data}}(x))^2 dx}_{\text{const}} + \frac{1}{2} \int p_{\text{data}}(x) (\nabla_x \log p_{\theta}(x))^2 dx \\ &\quad - \int p_{\text{data}}(x) \nabla_x \log p_{\theta}(x) \nabla_x \log p_{\text{data}}(x) dx. \end{aligned}$$

Estimating the score function of data

$$\begin{aligned}& - \int p_{\text{data}}(x) \nabla_x \log p_{\theta}(x) \nabla_x \log p_{\text{data}}(x) dx \\&= - \int \nabla_x \log p_{\theta}(x) \nabla_x p_{\text{data}}(x) dx \\&= - p_{\text{data}}(x) \nabla_x \log p_{\theta}(x) \Big|_{-\infty}^{\infty} + \int p_{\text{data}}(x) \nabla_x^2 \log p_{\theta}(x) dx \\&\stackrel{(i)}{=} \mathbb{E}_{p_{\text{data}}} [\nabla_x^2 \log p_{\theta}(x)],\end{aligned}$$

Estimating the score function of data

Given: $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}(\mathbf{x})$

Goal: $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Score Model: $\mathbf{s}_{\theta}(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^d \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$

Objective: How to compare two vector fields of scores?

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x})\|_2^2]$$

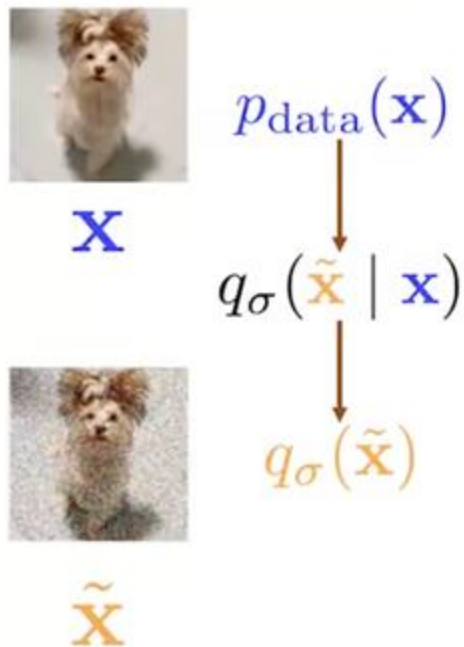
(Fisher divergence)

Score Matching [Hyvärinen 2005]:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \|\mathbf{s}_{\theta}(\mathbf{x})\|_2^2 + \text{trace} \left(\underbrace{\nabla_{\mathbf{x}} \mathbf{s}_{\theta}(\mathbf{x})}_{\text{Jacobian of } \mathbf{s}_{\theta}(\mathbf{x})} \right) \right]$$

Integration by parts
(Gauss's theorem)

Denoising score matching



- Matching the score of a noise-perturbed distribution

$$\frac{1}{2} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}})} [\|\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}}) - \mathbf{s}_{\theta}(\tilde{\mathbf{x}})\|_2^2]$$

- Denoising score matching (Vincent 2011)

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x})} [\|\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) - \mathbf{s}_{\theta}(\tilde{\mathbf{x}})\|_2^2]$$

↓
Scalable!

Denoising score matching

$$q_{\theta}(\tilde{x}) = \int q_{\theta}(\tilde{x} | x) p_{\text{data}}(x) dx$$

$$E_{q_{\theta}(\tilde{x})} \| \nabla_{\tilde{x}} \log q_{\theta}(\tilde{x}) - s_{\theta}(\tilde{x}) \|_2^2$$

Denoising score matching

$$\begin{aligned} & \int q_{\theta}(\tilde{x}) \left(\nabla_{\tilde{x}} \log q_{\theta}(\tilde{x}) - s_{\theta}(\tilde{x}) \right)^2 d\tilde{x} = \\ &= \int q_{\theta}(\tilde{x}) s_{\theta}^2(\tilde{x}) d\tilde{x} - 2 \int q_{\theta}(\tilde{x}) s_{\theta}(\tilde{x}) \nabla_{\tilde{x}} \log q_{\theta}(\tilde{x}) d\tilde{x} + \\ &+ \underbrace{\int q_{\theta}(\tilde{x}) \left(\nabla_{\tilde{x}} \log q_{\theta}(\tilde{x}) \right)^2 d\tilde{x}}_{\text{const}} \end{aligned}$$

Denoising score matching

$$\begin{aligned}\int q_{\theta}(\hat{x}) s_{\theta}^2(\hat{x}) d\hat{x} &= \int \int p_{\text{data}}(x) q_{\theta}(\hat{x}|x) dx s_{\theta}^2(\hat{x}) d\hat{x} \\&= \int p_{\text{data}}(x) \left(\int s_{\theta}^2(\hat{x}) q_{\theta}(\hat{x}|x) d\hat{x} \right) dx = \\&= E_{p_{\text{data}}(x)} E_{q_{\theta}(\hat{x}|x)} s_{\theta}^2(\hat{x})\end{aligned}$$

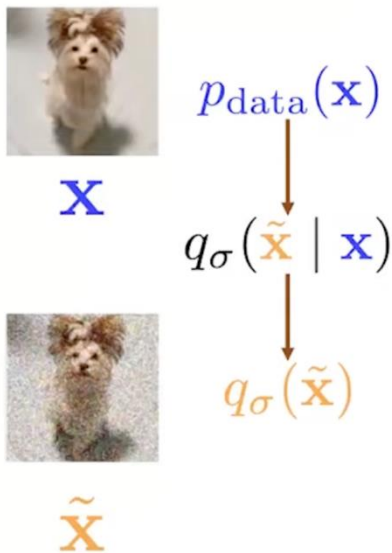
Denoising score matching

$$\begin{aligned}
 \int q_{\sigma}(\tilde{x}) s_{\theta}(\tilde{x}) \nabla_{\tilde{x}} \log q_{\sigma}(\tilde{x}) d\tilde{x} &= \int \nabla_{\tilde{x}} \log q_{\sigma}(\tilde{x}) = \frac{\nabla q_{\sigma}(\tilde{x})}{q_{\sigma}(\tilde{x})} d\tilde{x} = \\
 &= \int s_{\theta}(\tilde{x}) \nabla_{\tilde{x}} q_{\sigma}(\tilde{x}) d\tilde{x} = \int s_{\theta}(\tilde{x}) \nabla_{\tilde{x}} \left(\int p_{data}(x) q_{\sigma}(\tilde{x}|x) dx \right) d\tilde{x} = \\
 &= \int s_{\theta}(\tilde{x}) \int p_{data}(x) \nabla_{\tilde{x}} q_{\sigma}(\tilde{x}|x) dx d\tilde{x} = \\
 &= \int p_{data}(x) \int s_{\theta}(\tilde{x}) \nabla_{\tilde{x}} q_{\sigma}(\tilde{x}|x) d\tilde{x} dx =
 \end{aligned}$$

Denoising score matching

$$\begin{aligned}
 &= \int p_{data}(x) \int q_{\sigma}(\hat{x}|x) s_{\theta}(\hat{x}) \nabla_{\hat{x}} \log q_{\sigma}(\hat{x}|x) d\hat{x} dx = \\
 &= E_{p_{data}(x)} E_{q_{\sigma}(\hat{x}|x)} [s_{\theta}(\hat{x}) \nabla_{\hat{x}} \log q_{\sigma}(\hat{x}|x)]
 \end{aligned}$$

Denoising score matching



- Matching the score of a noise-perturbed distribution

$$\frac{1}{2} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})} [\|\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}) - \mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2]$$

- Denoising score matching (Vincent 2011)

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})} [\|\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}} | \mathbf{x}) - \mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2]$$

Scalable!

Denoising score matching

$$\begin{aligned}\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x}) &= \nabla_{\tilde{\mathbf{x}}} \log \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I}) \\&= \nabla_{\tilde{\mathbf{x}}} \log \frac{\exp(-\frac{1}{2}(\tilde{\mathbf{x}} - \mathbf{x})^T \cdot (\sigma^2 \cdot \mathbf{I})^{-1} \cdot (\tilde{\mathbf{x}} - \mathbf{x}))}{\sqrt{(2\pi)^d |\sigma^2 \cdot \mathbf{I}|}} \\&= \nabla_{\tilde{\mathbf{x}}} \log \exp(-\frac{1}{2}(\tilde{\mathbf{x}} - \mathbf{x})^T \cdot (\sigma^2 \cdot \mathbf{I})^{-1} \cdot (\tilde{\mathbf{x}} - \mathbf{x})) \\&\quad - \underbrace{\nabla_{\tilde{\mathbf{x}}} \log \sqrt{(2\pi)^d |\sigma^2 \cdot \mathbf{I}|}}_{=0} \\&= -\frac{1}{2\sigma^2} \nabla_{\tilde{\mathbf{x}}} (\tilde{\mathbf{x}} - \mathbf{x})^T \cdot \mathbf{I} \cdot (\tilde{\mathbf{x}} - \mathbf{x}) \\&= -\frac{1}{2\sigma^2} \nabla_{\tilde{\mathbf{x}}} (\tilde{\mathbf{x}} - \mathbf{x})^2 \\&= -\frac{(\tilde{\mathbf{x}} - \mathbf{x})}{\sigma^2} = \frac{(\mathbf{x} - \tilde{\mathbf{x}})}{\sigma^2}.\end{aligned}$$

Denoising score matching

- Sample a minibatch of datapoints $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\} \sim p_{\text{data}}(\mathbf{x})$
- Sample a minibatch of perturbed datapoints $\{\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \dots, \tilde{\mathbf{x}}_n\} \sim q_{\sigma}(\tilde{\mathbf{x}})$
 $\tilde{\mathbf{x}}_i \sim q_{\sigma}(\tilde{\mathbf{x}}_i \mid \mathbf{x}_i)$

- Estimate the denoising score matching loss with empirical means

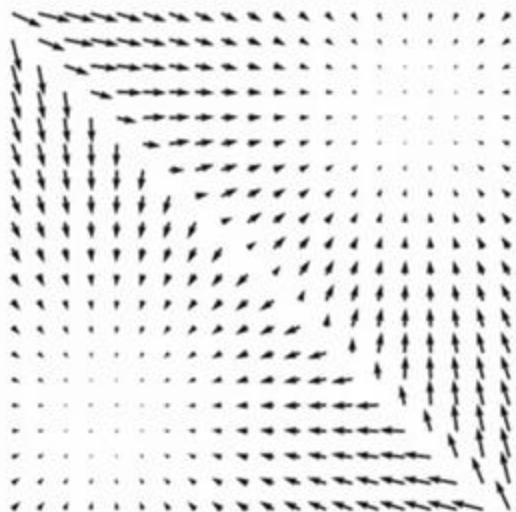
$$\frac{1}{2n} \sum_{i=1}^n [\|s_{\theta}(\tilde{\mathbf{x}}_i) - \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}}_i \mid \mathbf{x}_i)\|_2^2]$$

- If Gaussian perturbation

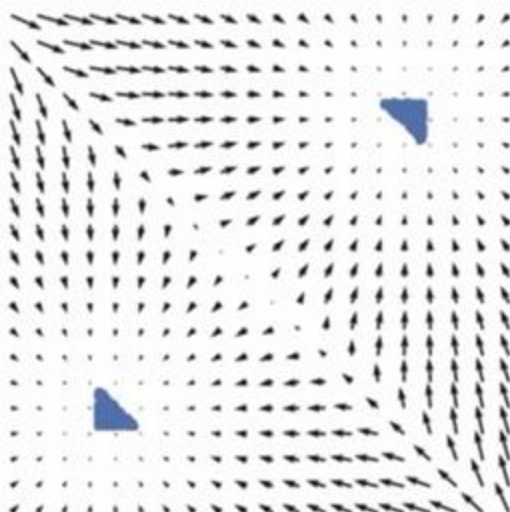
$$\frac{1}{2n} \sum_{i=1}^n \left[\left\| s_{\theta}(\tilde{\mathbf{x}}_i) + \frac{\tilde{\mathbf{x}}_i - \mathbf{x}_i}{\sigma^2} \right\|_2^2 \right]$$

- Stochastic gradient descent
- Need to choose a very small σ !

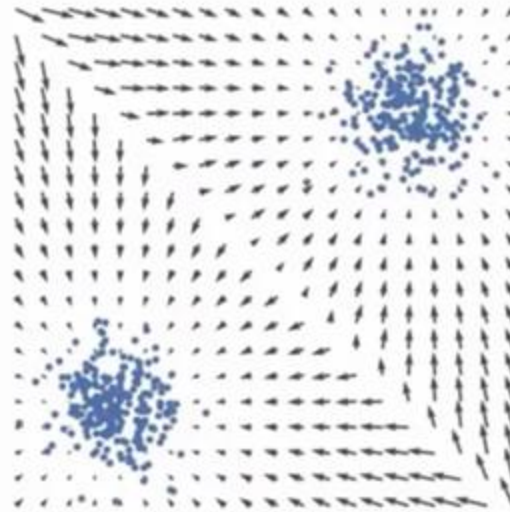
Sampling from score function



Score function



Follow the scores

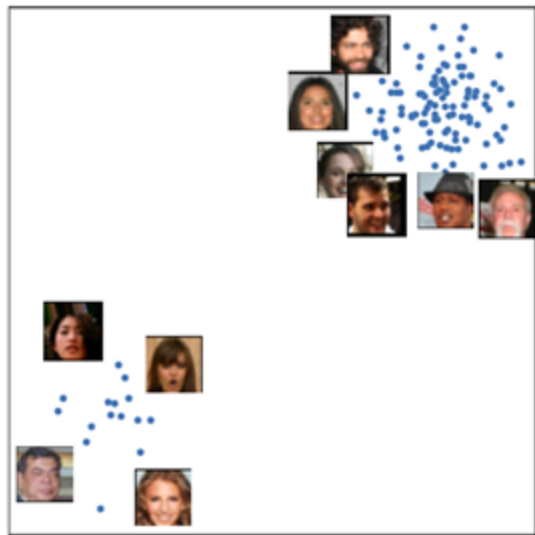


Follow the noisy scores

$$\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + \epsilon \nabla_{\mathbf{x}} \log p(\mathbf{x}) \quad \mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + \epsilon \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \sqrt{2\epsilon} \mathbf{z}_i$$

Langevin dynamics

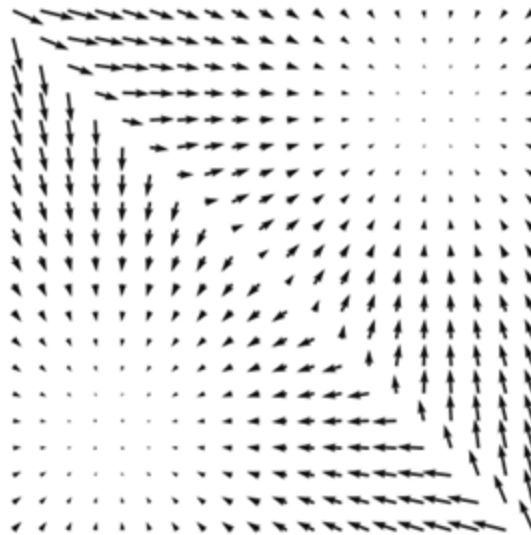
Sampling from score function



Data samples

$$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p(\mathbf{x})$$

score
matching



Scores

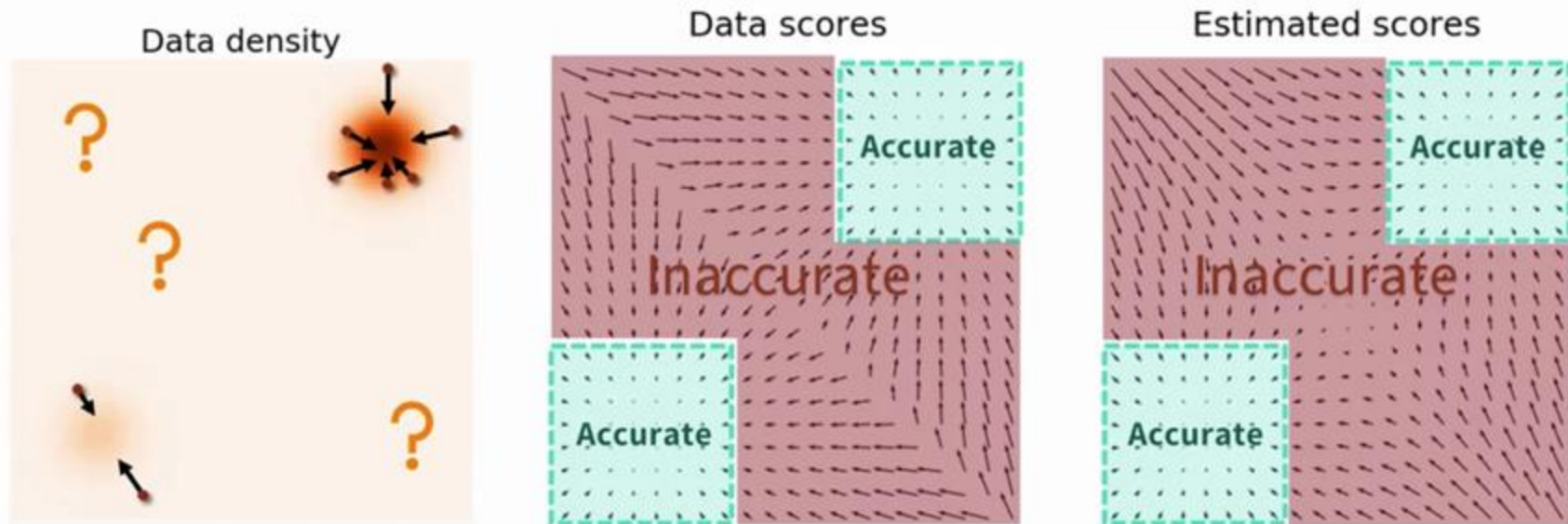
$$\mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Langevin
dynamics



New samples

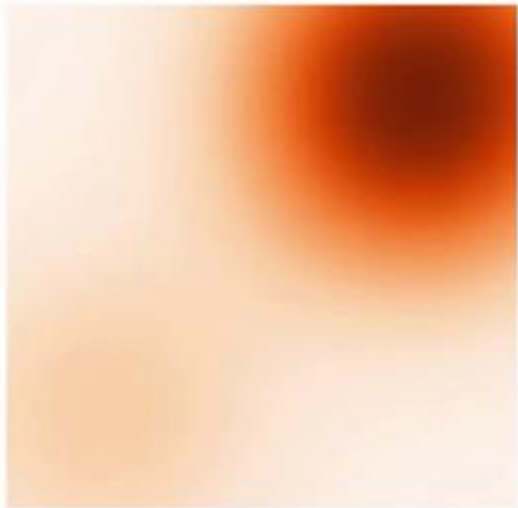
Challenge in low data regions



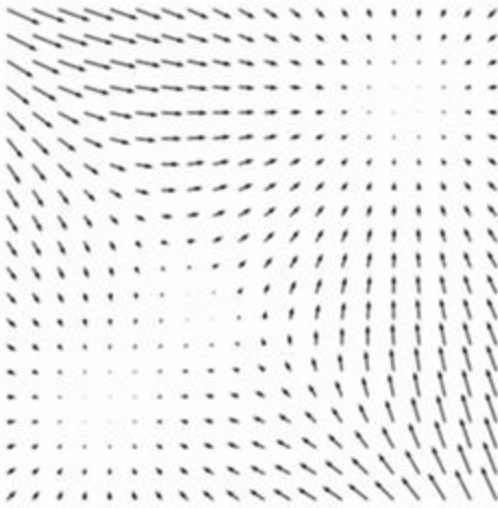
$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x})\|_2^2]$$

Challenge in low data regions

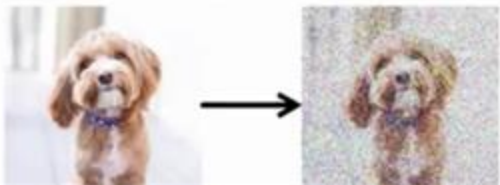
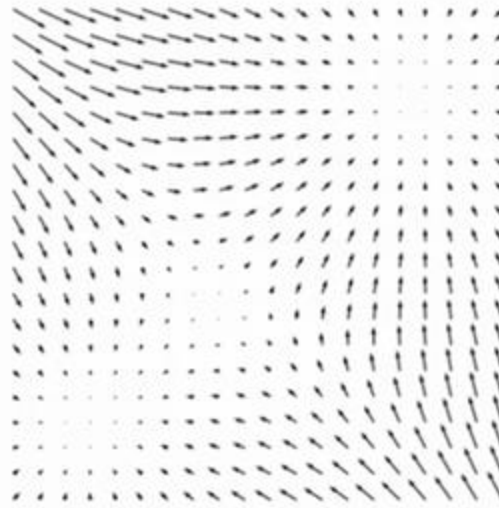
Perturbed density



Perturbed scores



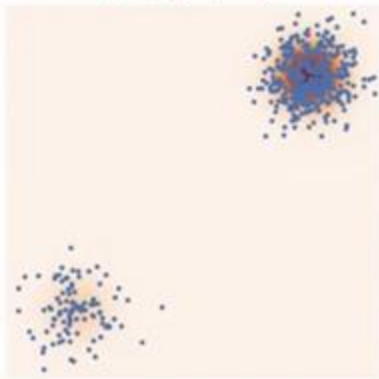
Estimated scores



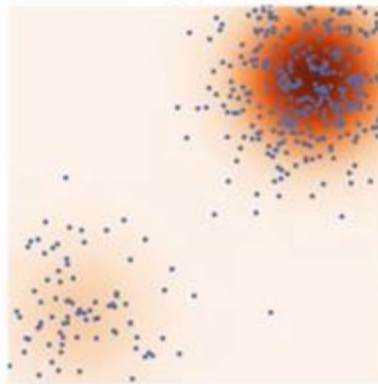
Using multiple noise levels

Data

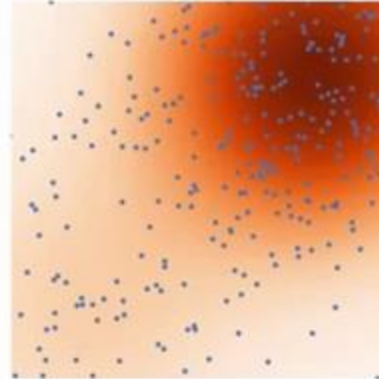
$$p_{\sigma_1}(\mathbf{x})$$



$$p_{\sigma_2}(\mathbf{x})$$



$$p_{\sigma_3}(\mathbf{x})$$



Using multiple noise levels

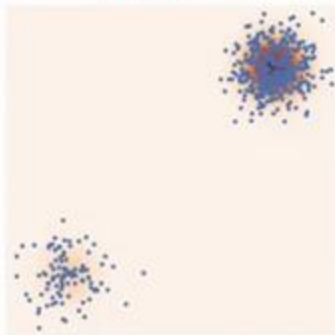
Data



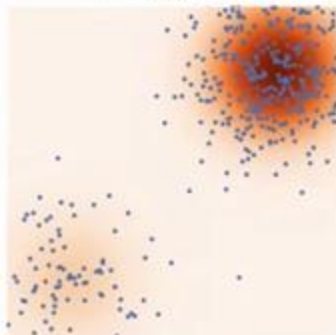
Using multiple noise levels

Data

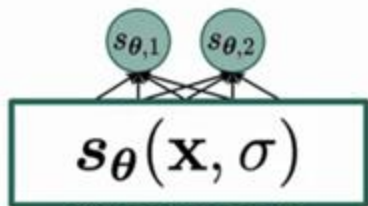
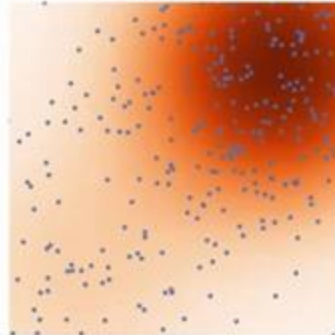
$p_{\sigma_1}(\mathbf{x})$



$p_{\sigma_2}(\mathbf{x})$



$p_{\sigma_3}(\mathbf{x})$



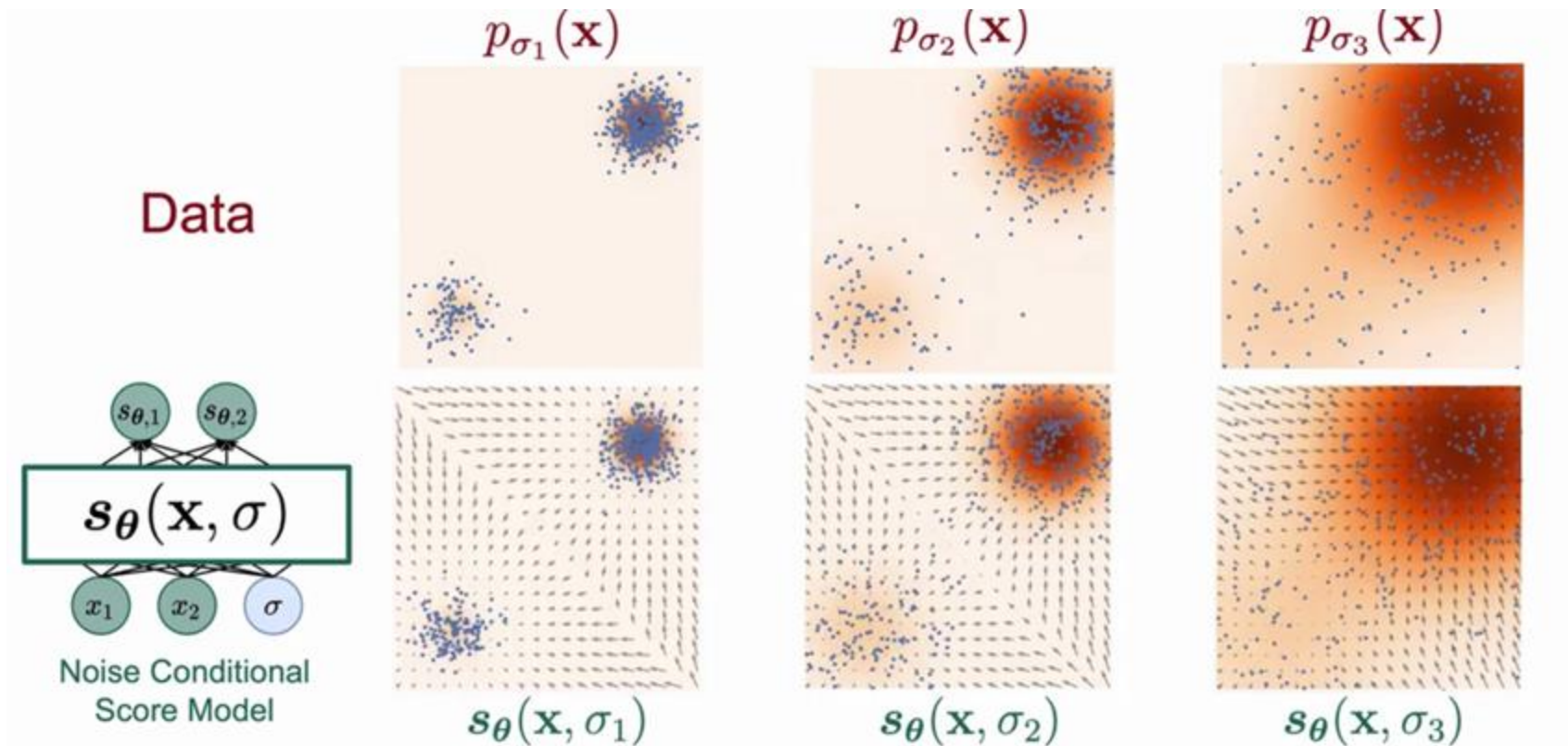
Noise Conditional
Score Model

Positive weighting
function

$$\frac{1}{N} \sum_{i=1}^N \lambda(\sigma_i) \underbrace{\mathbb{E}_{p_{\sigma_i}(\mathbf{x})} [\|\nabla_{\mathbf{x}} \log p_{\sigma_i}(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x}, \sigma_i)\|_2^2]}_{\text{Score matching loss}}$$

Score matching loss

Using multiple noise levels



Learning NCSNs via score matching

$$\ell(\boldsymbol{\theta}; \sigma) \triangleq \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\tilde{\mathbf{x}} \sim \mathcal{N}(\mathbf{x}, \sigma^2 I)} \left[\left\| \mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}, \sigma) + \frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2} \right\|_2^2 \right]$$

$$\mathcal{L}(\boldsymbol{\theta}; \{\sigma_i\}_{i=1}^L) \triangleq \frac{1}{L} \sum_{i=1}^L \lambda(\sigma_i) \ell(\boldsymbol{\theta}; \sigma_i)$$

Algorithm 1 Annealed Langevin dynamics.

Require: $\{\sigma_i\}_{i=1}^L, \epsilon, T$.

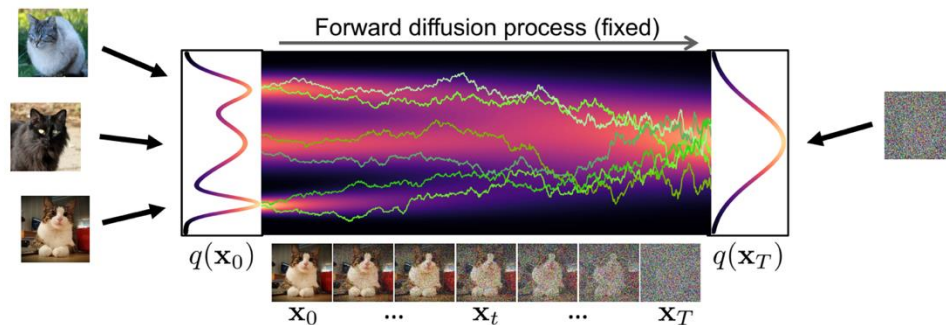
- 1: Initialize $\tilde{\mathbf{x}}_0$
 - 2: **for** $i \leftarrow 1$ to L **do**
 - 3: $\alpha_i \leftarrow \epsilon \cdot \sigma_i^2 / \sigma_L^2$ $\triangleright \alpha_i$ is the step size.
 - 4: **for** $t \leftarrow 1$ to T **do**
 - 5: Draw $\mathbf{z}_t \sim \mathcal{N}(0, I)$
 - 6: $\tilde{\mathbf{x}}_t \leftarrow \tilde{\mathbf{x}}_{t-1} + \frac{\alpha_i}{2} \mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_{t-1}, \sigma_i) + \sqrt{\alpha_i} \mathbf{z}_t$
 - 7: **end for**
 - 8: $\tilde{\mathbf{x}}_0 \leftarrow \tilde{\mathbf{x}}_T$
 - 9: **end for**
 - return** $\tilde{\mathbf{x}}_T$
-

Score matching perspective

- Estimating scores from data instead of density
- Apply score matching techniques
- Utilize denoising score matching
- Use multiple noise levels

SDE Perspective

The Generative Reverse Stochastic Differential Equation



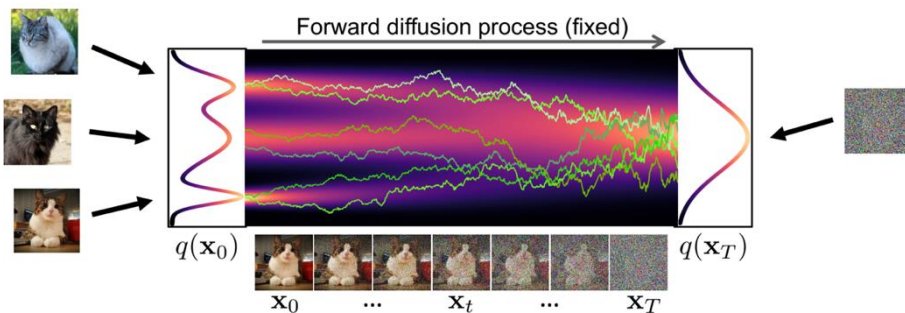
Forward Diffusion SDE:

$$dx_t = -\frac{1}{2}\beta(t)x_t dt + \sqrt{\beta(t)} d\omega_t$$

**Reverse Generative
Diffusion SDE:**

$$dx_t = \underbrace{\left[-\frac{1}{2}\beta(t)x_t - \beta(t) \underbrace{\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)}_{\text{"Score Function"}} \right]}_{\text{drift term}} dt + \underbrace{\sqrt{\beta(t)} d\bar{\omega}_t}_{\text{diffusion term}}$$

Score Matching

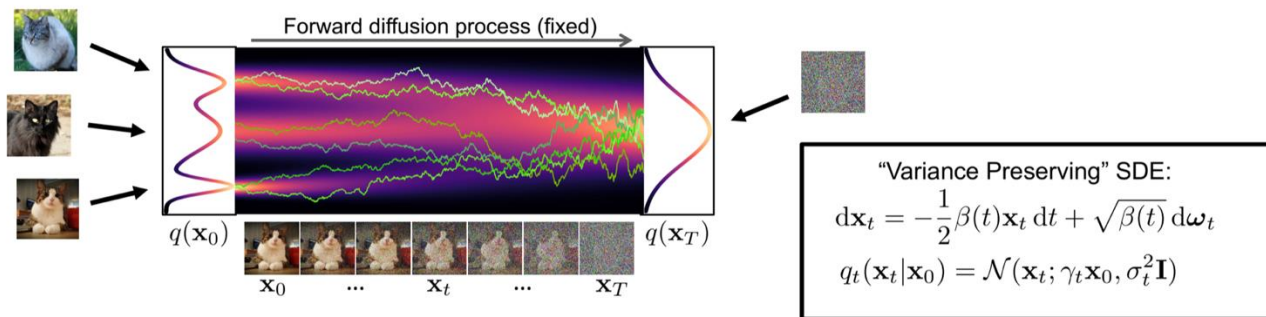


Naïve idea, learn model for the score function by direct regression?

$$\min_{\theta} \underbrace{\mathbb{E}_{t \sim \mathcal{U}(0, T)}}_{\text{diffusion time } t} \underbrace{\mathbb{E}_{\mathbf{x}_t \sim q_t(\mathbf{x}_t)}}_{\text{diffused data } \mathbf{x}_t} \underbrace{\|\mathbf{s}_{\theta}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)\|_2^2}_{\text{score of diffused data (marginal)}}$$

➡ But $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ (**score of the *marginal diffused density* $q_t(\mathbf{x}_t)$**) is not tractable!

Denoising Score Matching



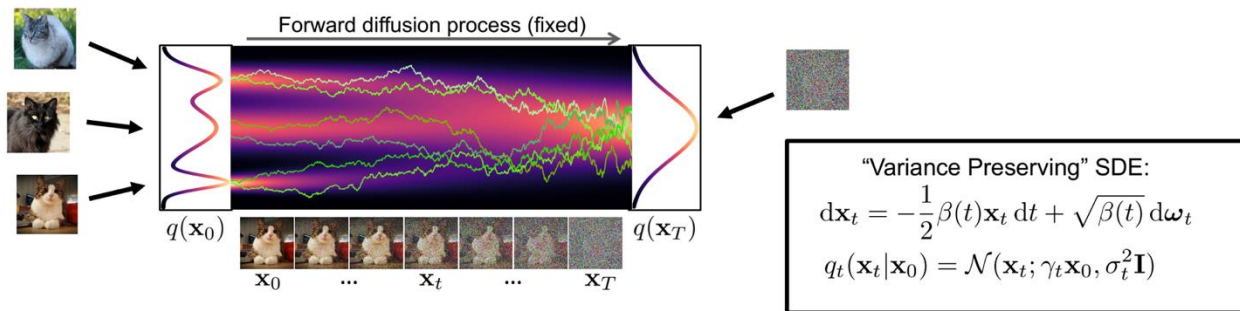
Instead, diffuse individual data points $\mathbf{x}_0$. Diffused $q_t(\mathbf{x}_t|\mathbf{x}_0)$ *is* tractable!

Denoising Score Matching:

$$\min_{\boldsymbol{\theta}} \underbrace{\mathbb{E}_{t \sim \mathcal{U}(0,T)}}_{\text{diffusion time } t} \underbrace{\mathbb{E}_{\mathbf{x}_0 \sim q_0(\mathbf{x}_0)}}_{\text{data sample } \mathbf{x}_0} \underbrace{\mathbb{E}_{\mathbf{x}_t \sim q_t(\mathbf{x}_t|\mathbf{x}_0)}}_{\text{diffused data sample } \mathbf{x}_t} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t|\mathbf{x}_0)\|_2^2}_{\text{score of diffused data sample}}$$

➔ **After expectations, $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$!**

Denoising Score Matching



Instead, diffuse individual data points $\mathbf{x}_0$. Diffused $q_t(\mathbf{x}_t|\mathbf{x}_0)$ *is* tractable!

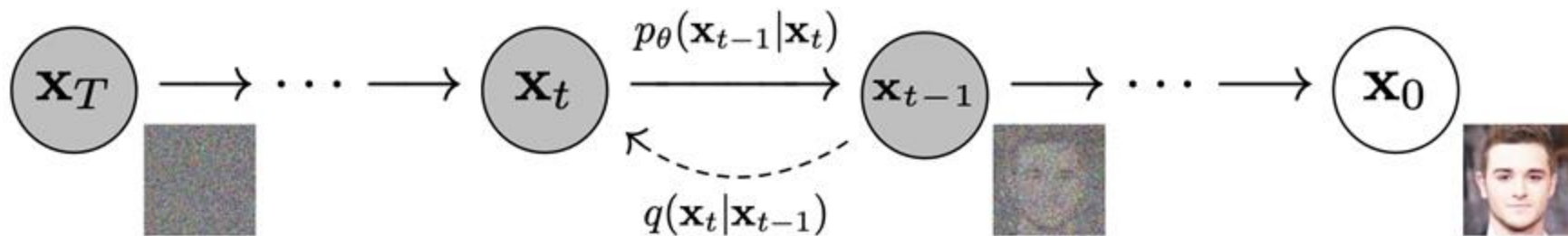
Denoising Score Matching:

$$\min_{\theta} \underbrace{\mathbb{E}_{t \sim \mathcal{U}(0, T)}}_{\text{diffusion time } t} \underbrace{\mathbb{E}_{\mathbf{x}_0 \sim q_0(\mathbf{x}_0)}}_{\text{data sample } \mathbf{x}_0} \underbrace{\mathbb{E}_{\mathbf{x}_t \sim q_t(\mathbf{x}_t|\mathbf{x}_0)}}_{\text{diffused data sample } \mathbf{x}_t} \underbrace{\|\mathbf{s}_{\theta}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t|\mathbf{x}_0)\|_2^2}_{\text{neural network score of diffused data sample}}$$

$$\rightarrow \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}(0, T)} \mathbb{E}_{\mathbf{x}_0 \sim q_0(\mathbf{x}_0)} \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \frac{1}{\sigma_t^2} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)\|_2^2$$

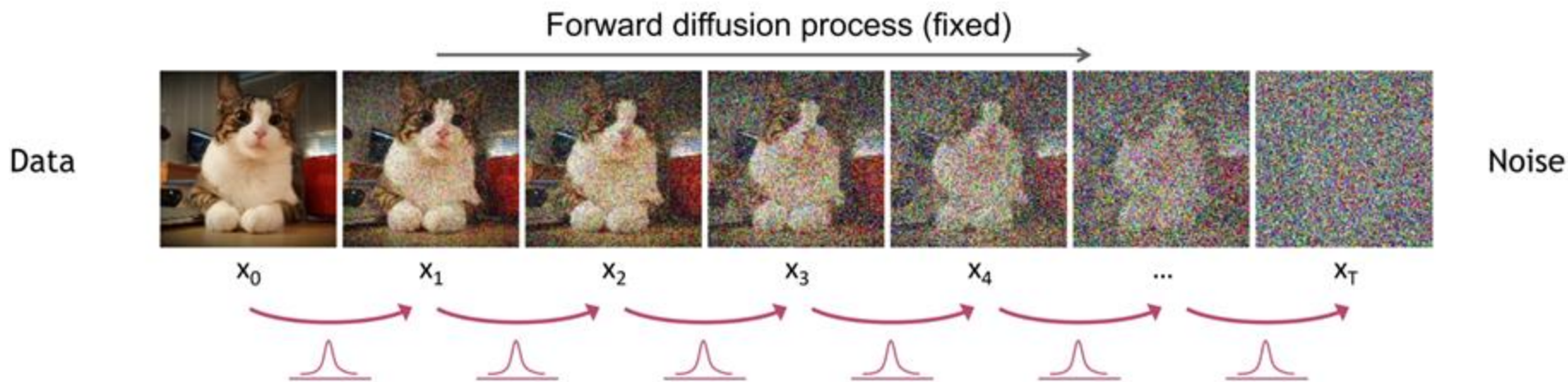
Latent Probabilistic Models Perspective

Denoising Diffusion Probabilistic Models

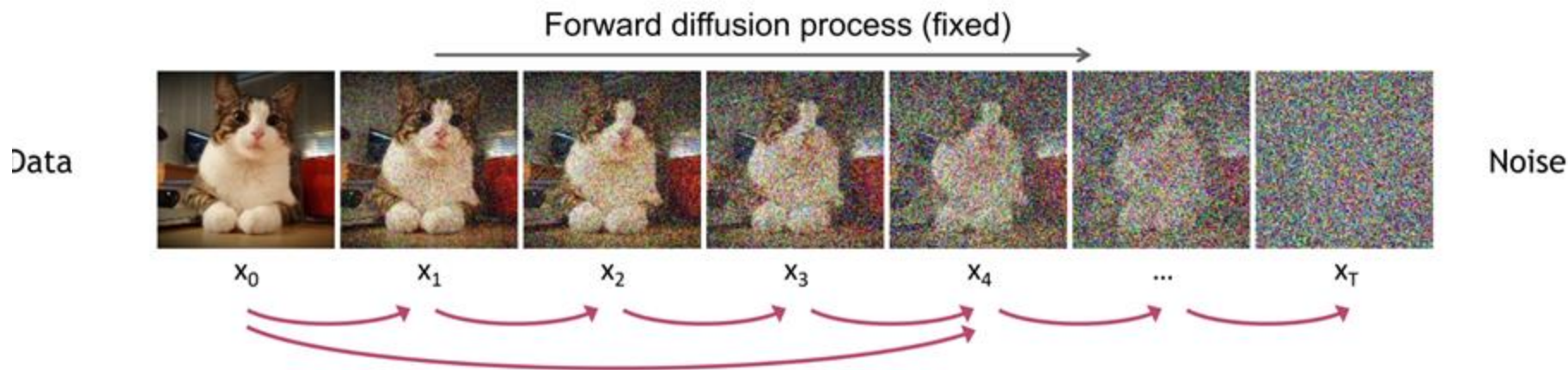


Denoising Diffusion Probabilistic Models

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$



Denoising Diffusion Probabilistic Models

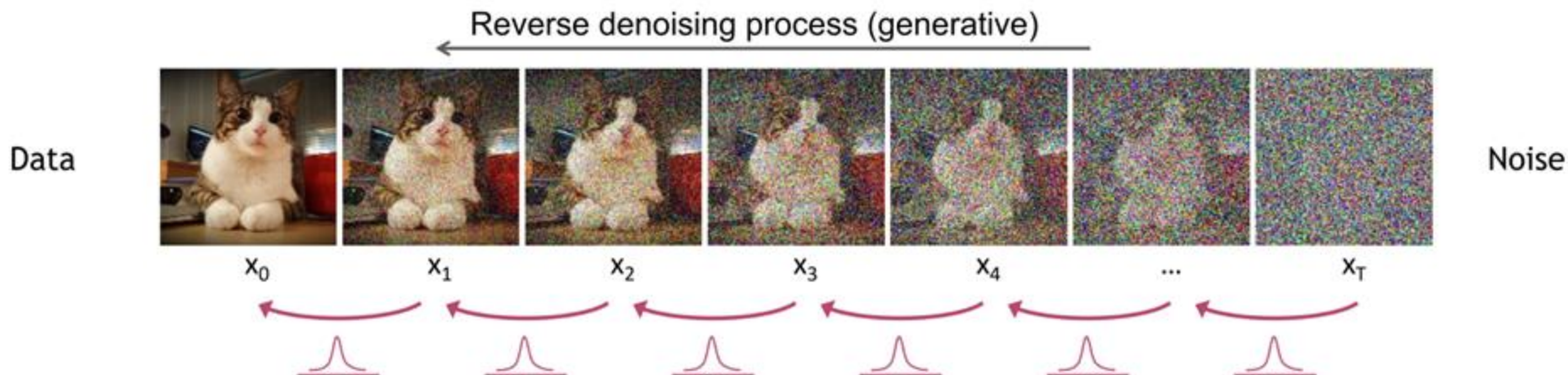


Define $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ $\rightarrow q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$ (Diffusion Kernel)

For sampling: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Denoising Diffusion Probabilistic Models

$$p_{\theta}(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$



Denoising Diffusion Probabilistic Models

$$\mathbb{E}[-\log p_{\theta}(\mathbf{x}_0)] \leq \mathbb{E}_q \left[-\log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] = \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] =: L$$

$$\mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t > 1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right]$$

Denoising Diffusion Probabilistic Models

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$\text{where } \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t \quad \text{and} \quad \tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t)\|^2 \right] + C$$

Denoising Diffusion Probabilistic Models

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t)\|^2 \right] + C$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \tilde{\boldsymbol{\mu}}_t \left(\mathbf{x}_t, \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t)) \right)$$

$$\mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t)\|^2 \right]$$

Denoising Diffusion Probabilistic Models

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
 $\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$
 - 6: **until** converged
-

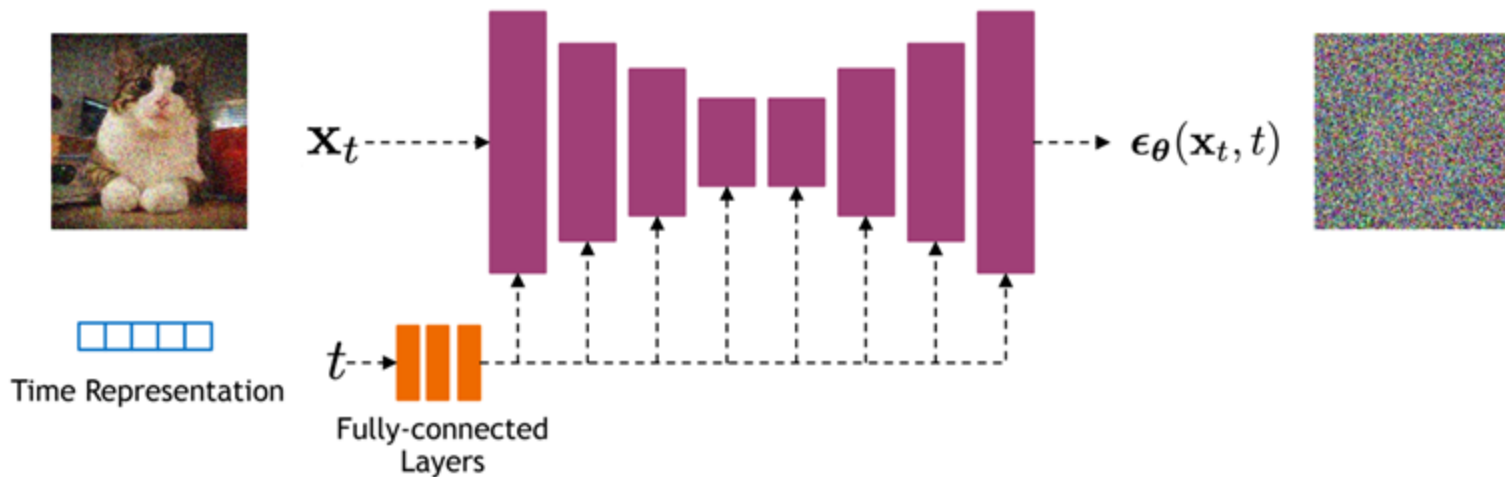
Algorithm 2 Sampling

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 2: **for** $t = T, \dots, 1$ **do**
 - 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
 - 4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
 - 5: **end for**
 - 6: **return** $\mathbf{x}_0$
-

Diffusion Models in CV

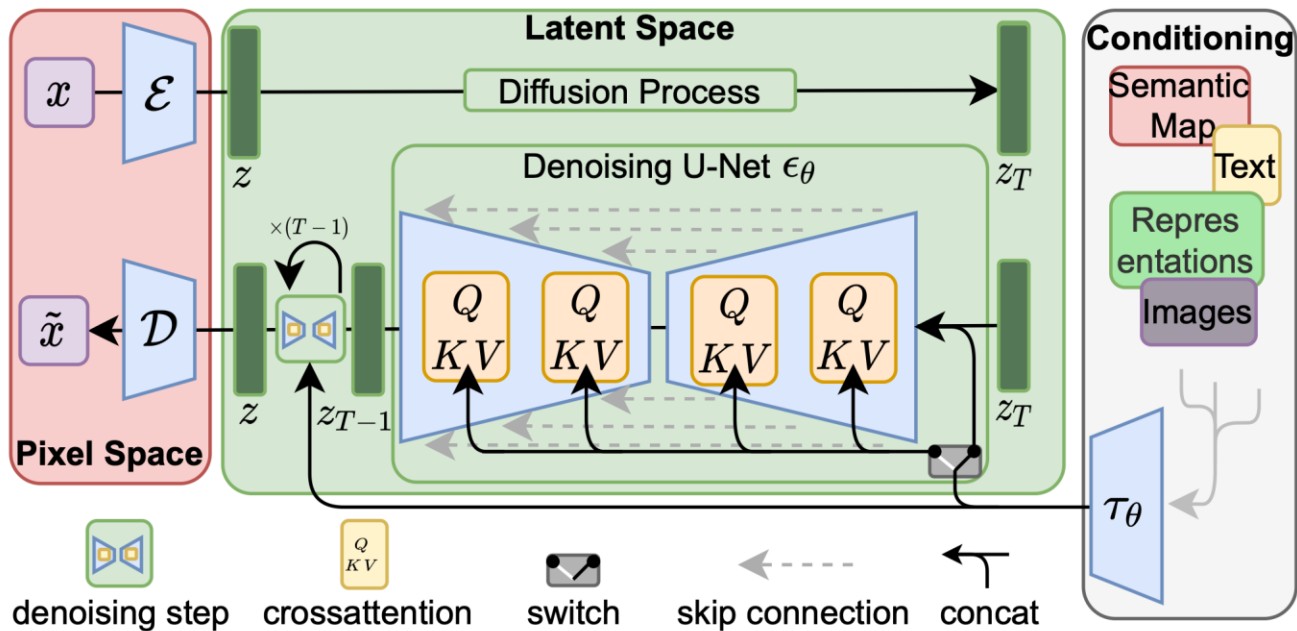
DDPM architecture

Diffusion models often use U-Net architectures with ResNet blocks and self-attention layers to represent $\epsilon_{\theta}(\mathbf{x}_t, t)$



Time representation: sinusoidal positional embeddings or random Fourier features.

StableDiffusion



Examples of Generated Images

Prompt:

“Ancient, mysterious temple in a mountain range, surrounded by misty clouds and tall peaks”



Sample from Stable Diffusion [1]

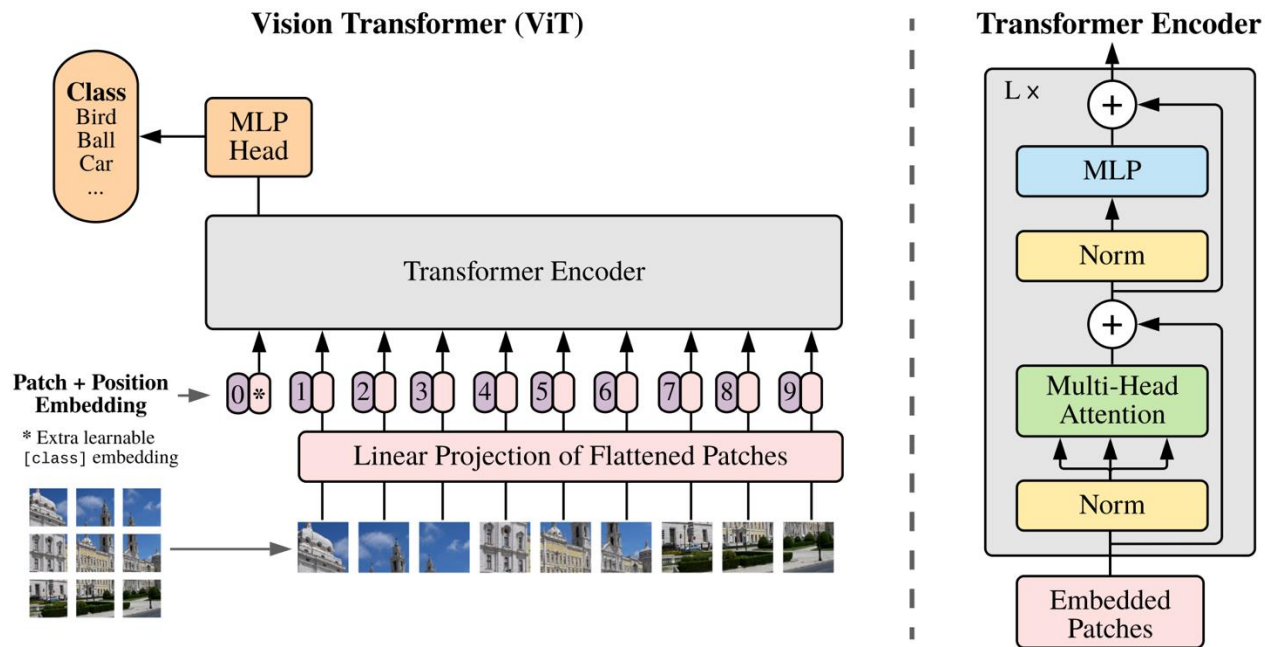
Prompt:

“A corgi’s head depicted as an explosion of a nebula”

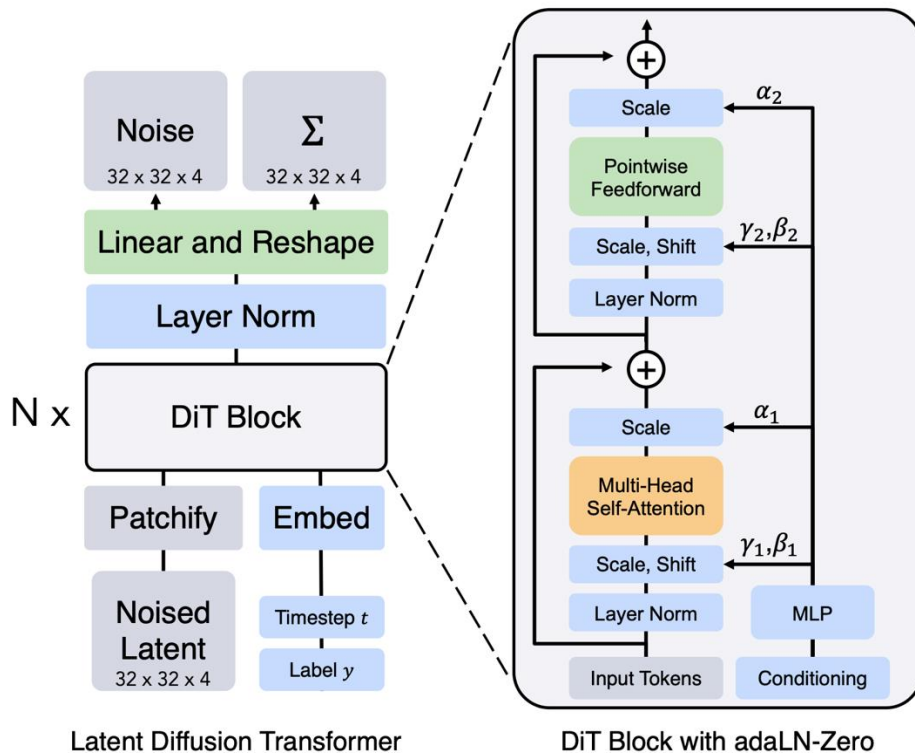


Sample from DALL-E 2 [2]

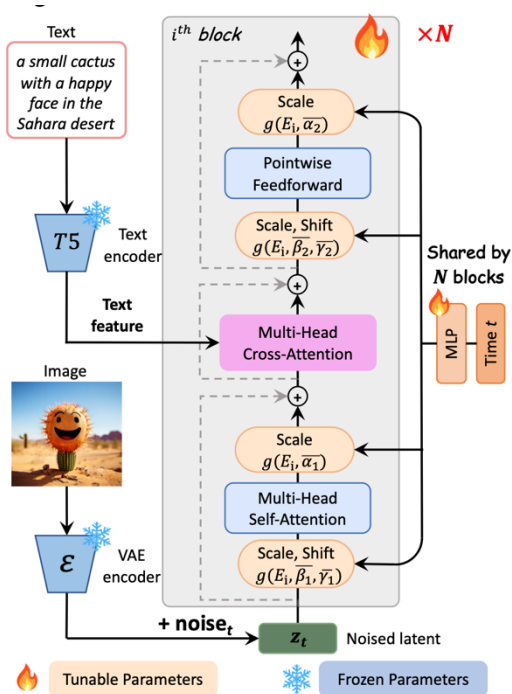
Vision Transformer (ViT)



Diffusion Transformer (DiT)



Pixart-alpha (text-to-image diffusion)

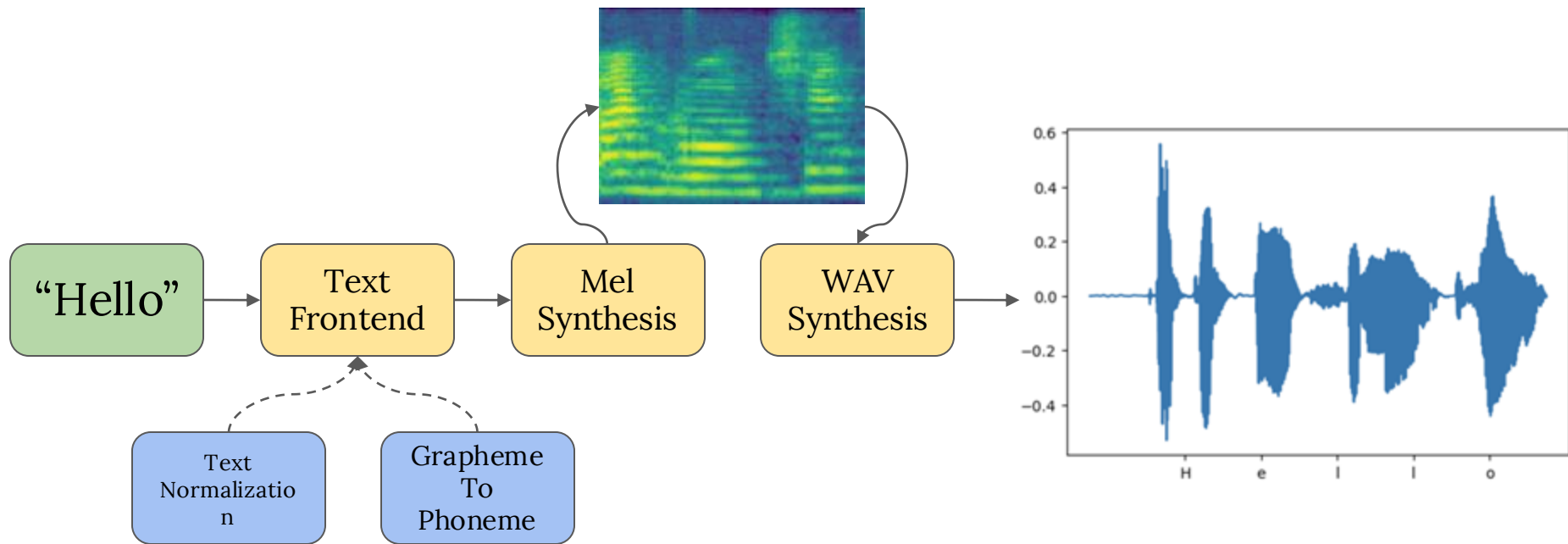


DiT models

- FLUX
- Stable Diffusion 3
- Sora 2
- Veo 3

Diffusion Models in Audio

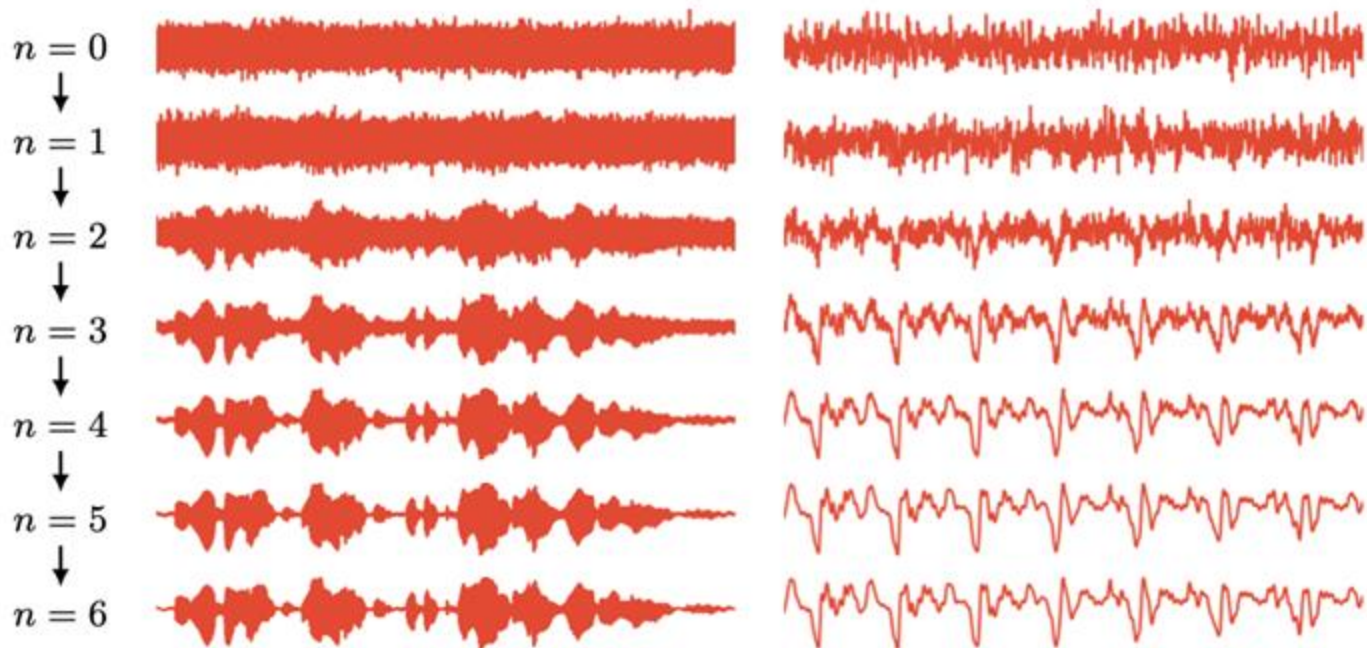
TTS recap



Diffusion Vocoders

- WaveGrad
- DiffWave

WaveGrad - Vocoder

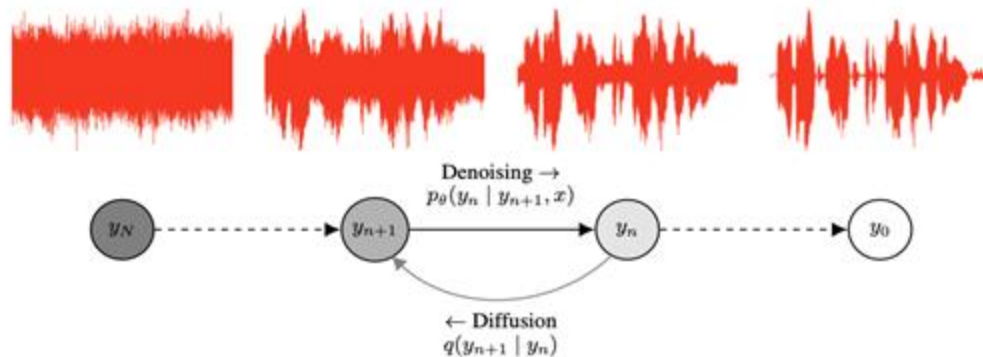


WaveGrad - Vocoder

$$s(y) = \nabla_y \log p(y).$$

$$\tilde{y}_{i+1} = \tilde{y}_i + \frac{\eta}{2} s(\tilde{y}_i) + \sqrt{\eta} z_i,$$

$$\mathbb{E}_{y \sim p(y)} \mathbb{E}_{\tilde{y} \sim q(\tilde{y} | y)} \left[\left\| s_{\theta}(\tilde{y}) - \nabla_{\tilde{y}} \log q(\tilde{y} | y) \right\|_2^2 \right]$$



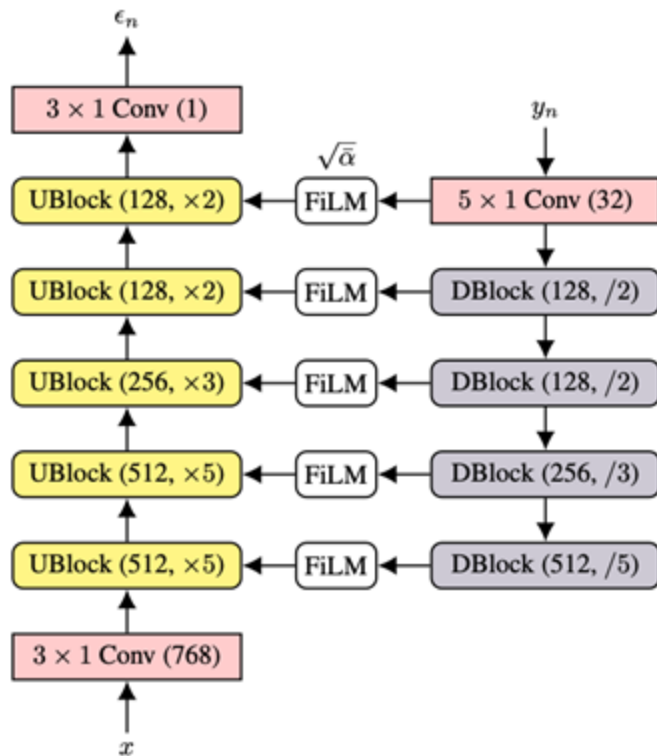
WaveGrad - Vocoder

$$\mathbb{E}_{\bar{\alpha}, \epsilon} \left[\left\| \epsilon_{\theta} \left(\sqrt{\bar{\alpha}} y_0 + \sqrt{1 - \bar{\alpha}} \epsilon, x, \sqrt{\bar{\alpha}} \right) - \epsilon \right\|_1 \right],$$

$$l_0 = 1, \quad l_s = \sqrt{\prod_{i=1}^s (1 - \beta_i)}.$$

We first sample a segment $s \sim U(\{1, \dots, S\})$, which provides a segment (l_{s-1}, l_s) , and then sample from this segment uniformly to give $\sqrt{\bar{\alpha}}$.

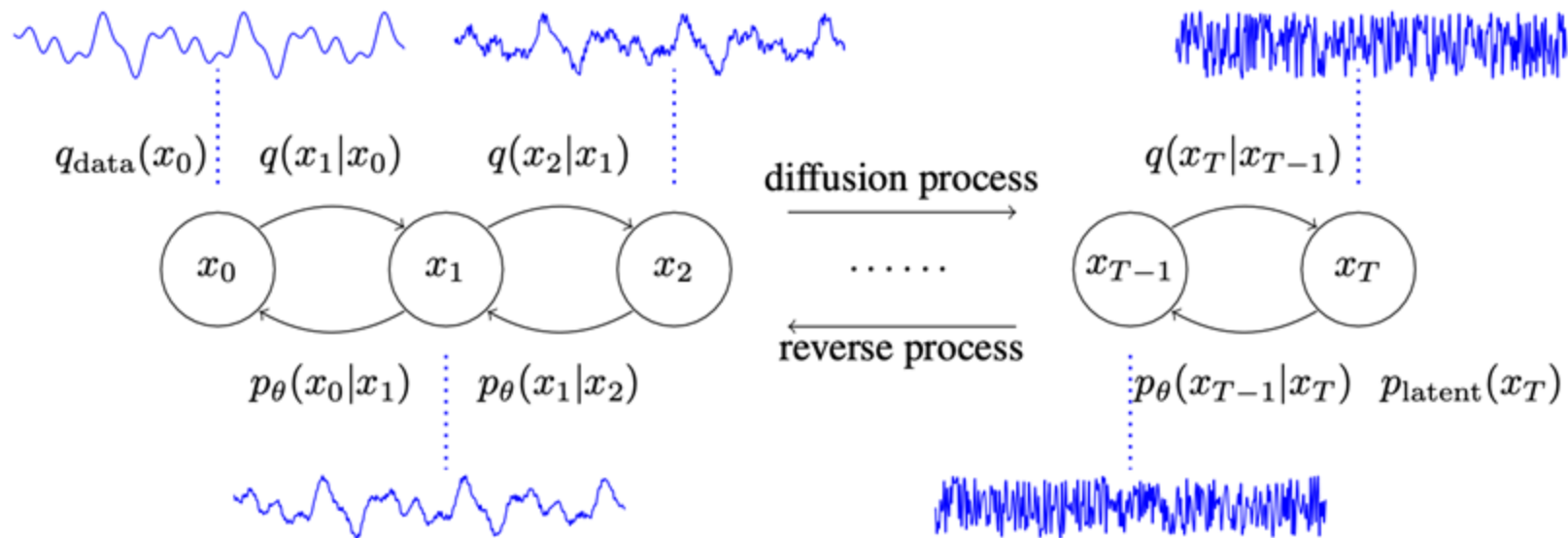
WaveGrad - Vocoder



WaveGrad - Vocoder

Model	MOS ($\uparrow$)
WaveRNN	4.49 ± 0.04
Parallel WaveGAN	3.92 ± 0.05
MelGAN	3.95 ± 0.06
Multi-band MelGAN	4.10 ± 0.05
GAN-TTS	4.34 ± 0.04
WaveGrad	
Base (6 iterations, continuous noise levels)	4.41 ± 0.03
Base (1,000 iterations, discrete indices)	4.47 ± 0.04
Large (1,000 iterations, discrete indices)	4.51 ± 0.04
Ground Truth	4.58 ± 0.05

DiffWave - Vocoder



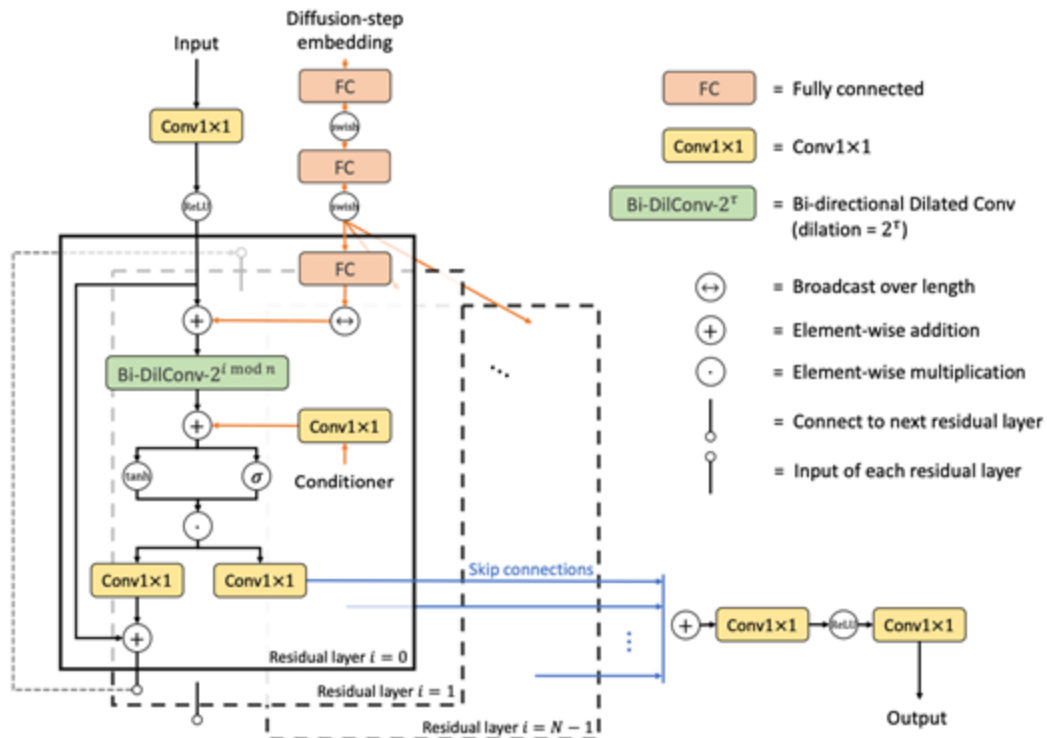
DiffWave - Vocoder

$$q(x_1, \dots, x_T | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}),$$

$$p_{\text{latent}}(x_T) = \mathcal{N}(0, I), \text{ and } p_{\theta}(x_0, \dots, x_{T-1} | x_T) = \prod_{t=1}^T p_{\theta}(x_{t-1} | x_t),$$

$$\min_{\theta} L_{\text{unweighted}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|_2^2$$

DiffWave - Vocoder



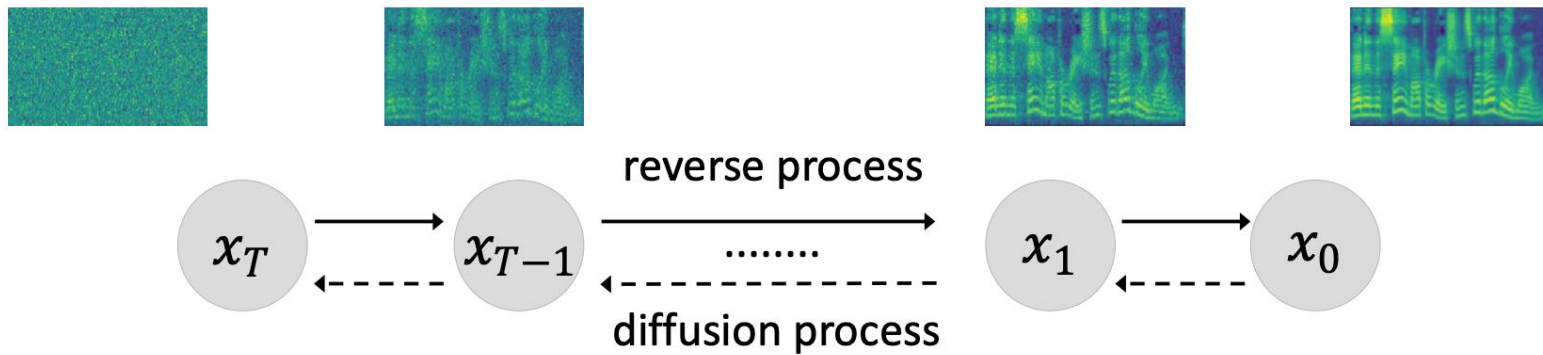
DiffWave - Vocoder

Model	T	T_{infer}	layers	res. channels	#param($\downarrow$)	MOS($\uparrow$)
WaveNet	—	—	30	128	4.57M	4.43 $\pm$ 0.10
ClariNet	—	—	60	64	2.17M	4.27 $\pm$ 0.09
WaveGlow	—	—	96	256	87.88M	4.33 $\pm$ 0.12
WaveFlow	—	—	64	64	5.91M	4.30 $\pm$ 0.11
WaveFlow	—	—	64	128	22.25M	4.40 $\pm$ 0.07
DiffWave _{BASE}	20	20	30	64	2.64M	4.31 $\pm$ 0.09
DiffWave _{BASE}	40	40	30	64	2.64M	4.35 $\pm$ 0.10
DiffWave _{BASE}	50	50	30	64	2.64M	4.38 $\pm$ 0.08
DiffWave _{LARGE}	200	200	30	128	6.91M	4.44 $\pm$ 0.07
DiffWave _{BASE} (Fast)	50	6	30	64	2.64M	4.37 $\pm$ 0.07
DiffWave _{LARGE} (Fast)	200	6	30	128	6.91M	4.42 $\pm$ 0.09
Ground-truth	—	—	—	—	—	4.52 $\pm$ 0.06

Diffusion TTS

- Diff-TTS
- GradTTS

Diff-TTS - Mel synthesis



Diff-TTS - Mel synthesis

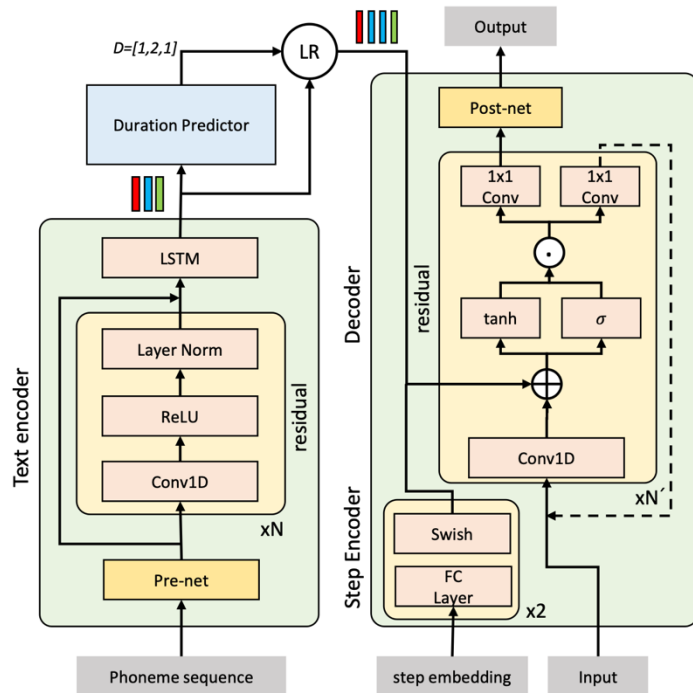
$$q(x_t|x_{t-1}, c) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I).$$

$$q(x_1 \dots, x_T|x_0, c) = \prod_{t=1}^T q(x_t|x_{t-1}).$$

$$p_\theta(x_0 \dots, x_{T-1}|x_T, c) = \prod_{t=1}^T p_\theta(x_{t-1}|x_t, c).$$

$$\min_{\theta} L(\theta) = \mathbb{E}_{x_0, \epsilon, t} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t, c)\|_1$$

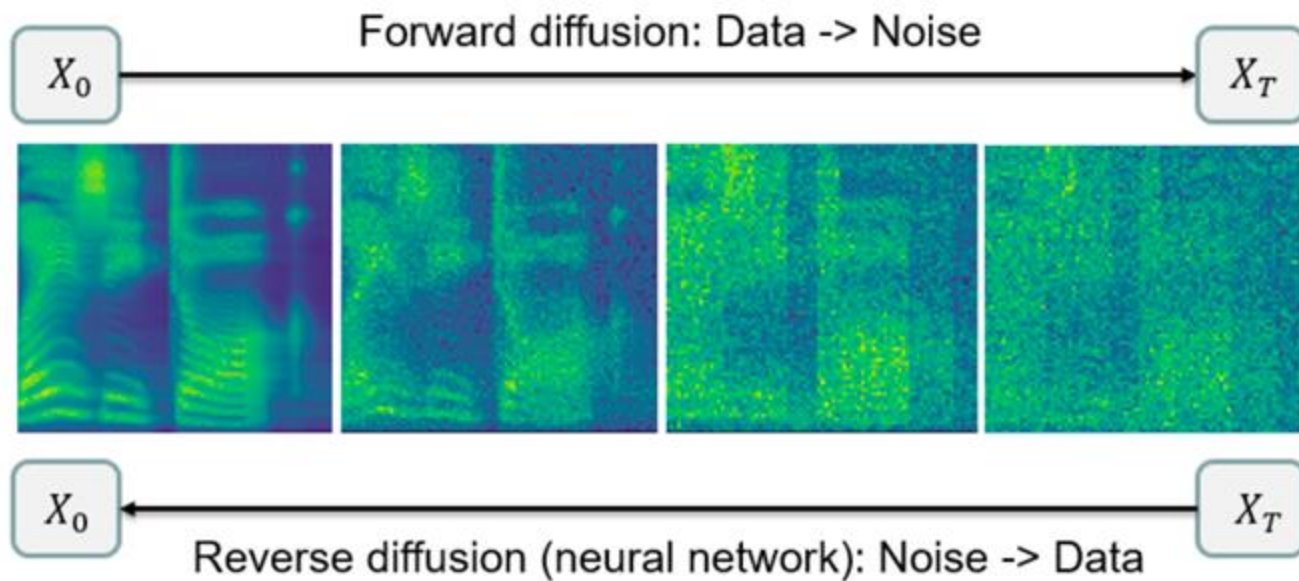
Diff-TTS - Mel synthesis



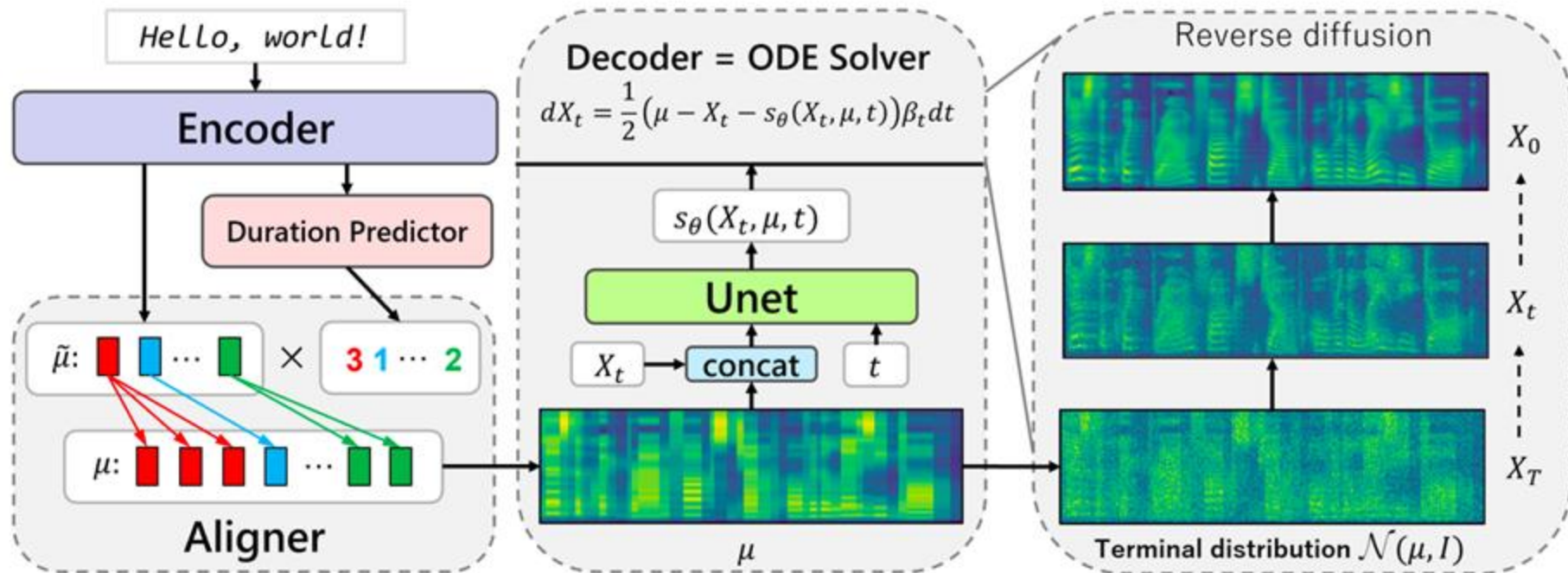
Diff-TTS - Mel synthesis

Method	5-scale MOS
GT	4.541 ± 0.057
GT(Mel + HiFiGAN)	4.323 ± 0.060
Tacotron2	4.006 ± 0.072
Glow-TTS	4.160 ± 0.070
Diff-TTS(T=400, $\gamma = 1$)	4.337 ± 0.064
Diff-TTS(T=400, $\gamma = 7$)	4.223 ± 0.066
Diff-TTS(T=400, $\gamma = 21$)	4.135 ± 0.070
Diff-TTS(T=400, $\gamma = 57$)	4.091 ± 0.067

GradTTS - Mel synthesis



GradTTS - Mel synthesis



GradTTS - Mel synthesis

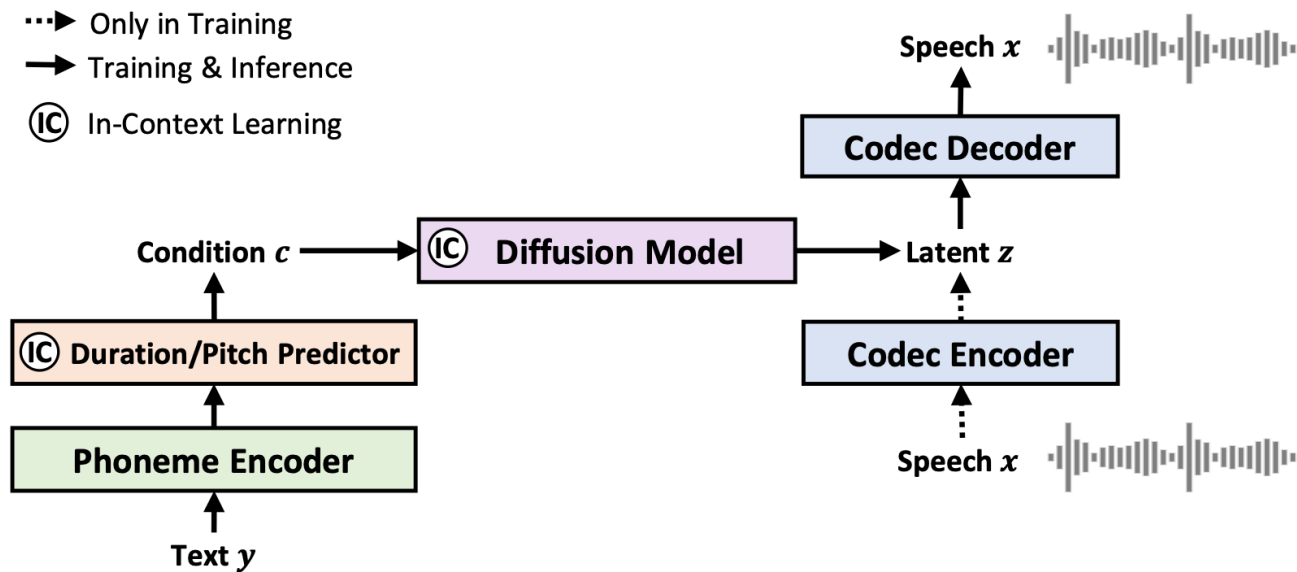
Table 2. Model comparison.

Model	Enc params ¹	Dec params	RTF	Log-likelihood	MOS
Grad-TTS-1000	<i>7.2m</i>	<i>7.6m</i>	3.663	0.174 ± 0.001	4.44 ± 0.05
Grad-TTS-100			0.363		4.38 ± 0.06
Grad-TTS-10			0.033		4.38 ± 0.06
Grad-TTS-4			0.012		3.96 ± 0.07
Glow-TTS	<i>7.2m</i>	<i>21.4m</i>	0.008	0.082	4.11 ± 0.07
FastSpeech	<i>24.5m</i>		0.004	—	3.68 ± 0.09
Tacotron2	<i>28.2m</i>		0.075	—	4.32 ± 0.07
Ground Truth	—		—	—	4.53 ± 0.06

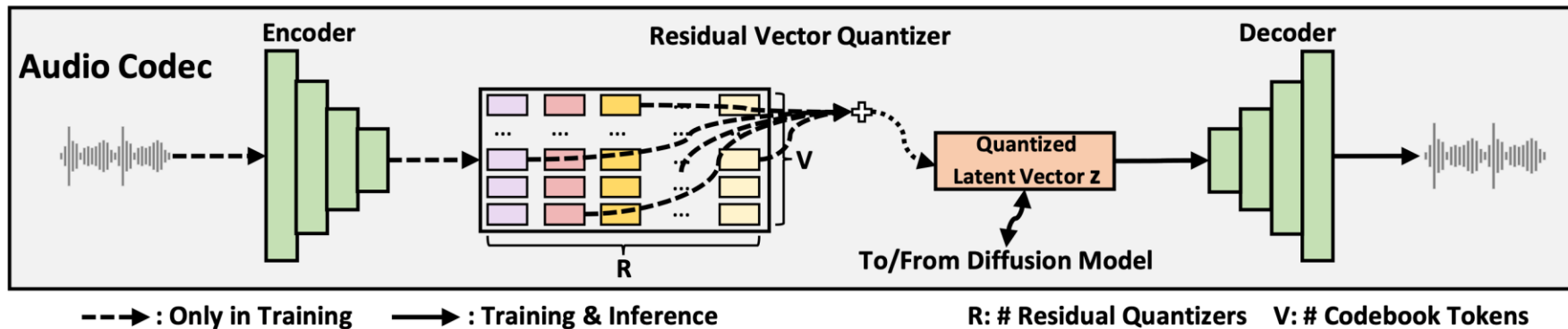
Latent diffusion TTS

- NaturalSpeech 2/3
- SimpleSpeech

NaturalSpeech 2



NaturalSpeech 2

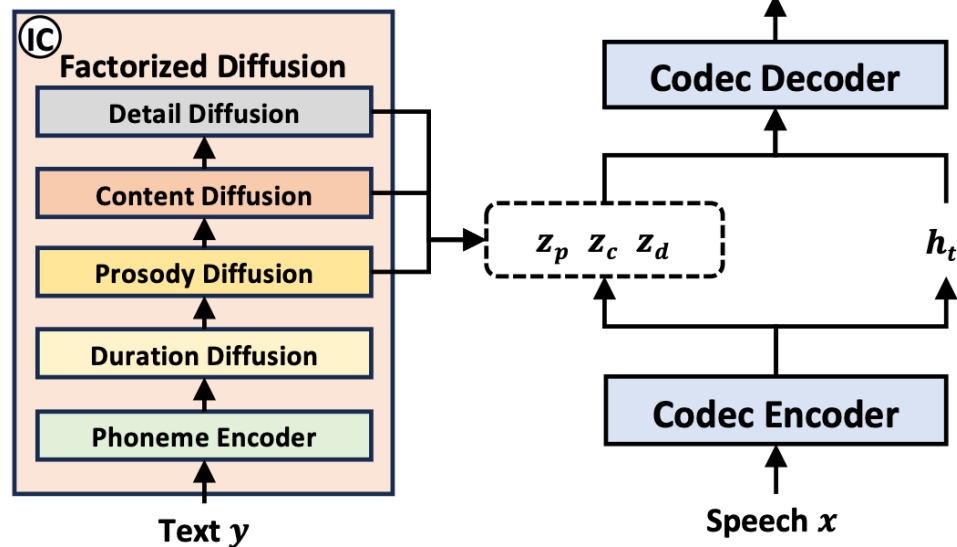


NaturalSpeech 2

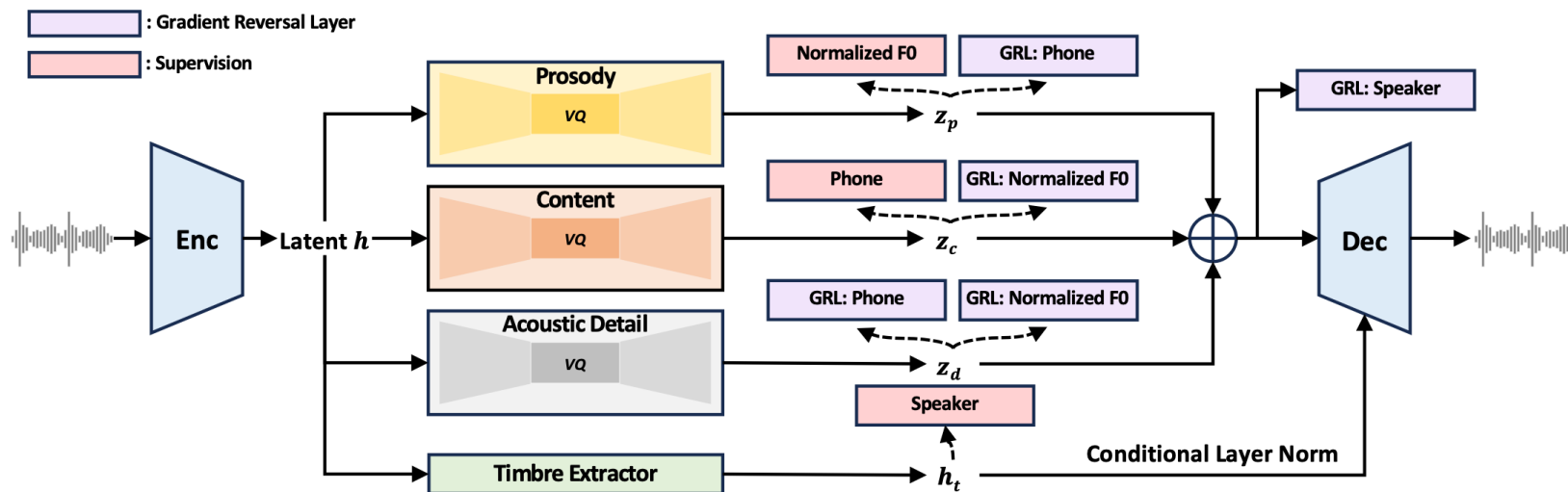
Dataset	LibriSpeech			VCTK		
Setting	CMOS↑	SMOS↑	WER↓	CMOS↑	SMOS↑	WER↓
Ground Truth	+0.04	4.27 $\pm$ 0.10	1.94	-0.21	4.05 $\pm$ 0.11	9.49
YourTTS (Casanova et al., 2022b)	-0.65	3.31 $\pm$ 0.09	7.10	-0.58	3.39 $\pm$ 0.08	14.80
FastSpeech 2 (Ren et al., 2021a)	-0.53	3.45 $\pm$ 0.08	2.10	-0.64	3.22 $\pm$ 0.10	8.26
FoundationTTS (Xue et al., 2023)	-0.32	3.81 $\pm$ 0.12	4.63	-0.39	3.42 $\pm$ 0.13	12.55
VALL-E (Wang et al., 2023)	-0.29	3.92 $\pm$ 0.11	5.72	-0.31	3.50 $\pm$ 0.10	14.68
NaturalSpeech 2	0.00	4.06 $\pm$ 0.11	2.01	0.00	3.62 $\pm$ 0.11	6.72
Comparison with demo cases						
VALL-E (Wang et al., 2023)*	-0.27	3.98 $\pm$ 0.12	-	-0.34	3.59 $\pm$ 0.13	-
NaturalSpeech 2	0.00	4.11 $\pm$ 0.11	-	0.00	3.71 $\pm$ 0.12	-
MegaTTS (Jiang et al., 2023b)*	-0.20	3.96 $\pm$ 0.09	-	-0.28	3.63 $\pm$ 0.08	-
NaturalSpeech 2	0.00	4.10 $\pm$ 0.11	-	0.00	3.74 $\pm$ 0.10	-
Voicebox (Le et al., 2023)*	-0.11	3.75 $\pm$ 0.11	-	-	-	-
NaturalSpeech 2	0.00	3.86 $\pm$ 0.11	-	-	-	-

NaturalSpeech 3

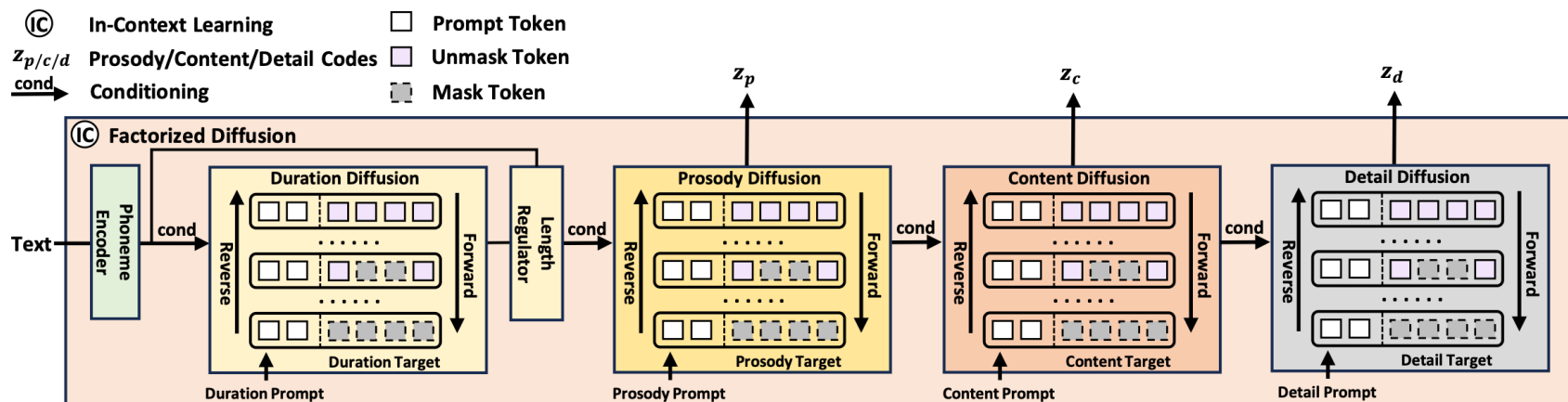
Ⓢ In-Context Learning
 $z_{p/c/d}$ Prosody/Content/Detail Codes
 h_t Timbre Embedding



NaturalSpeech 3



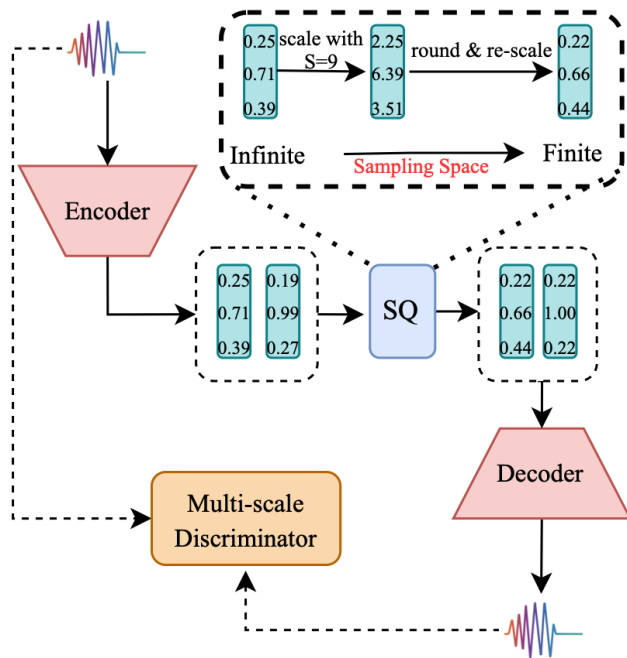
NaturalSpeech 3



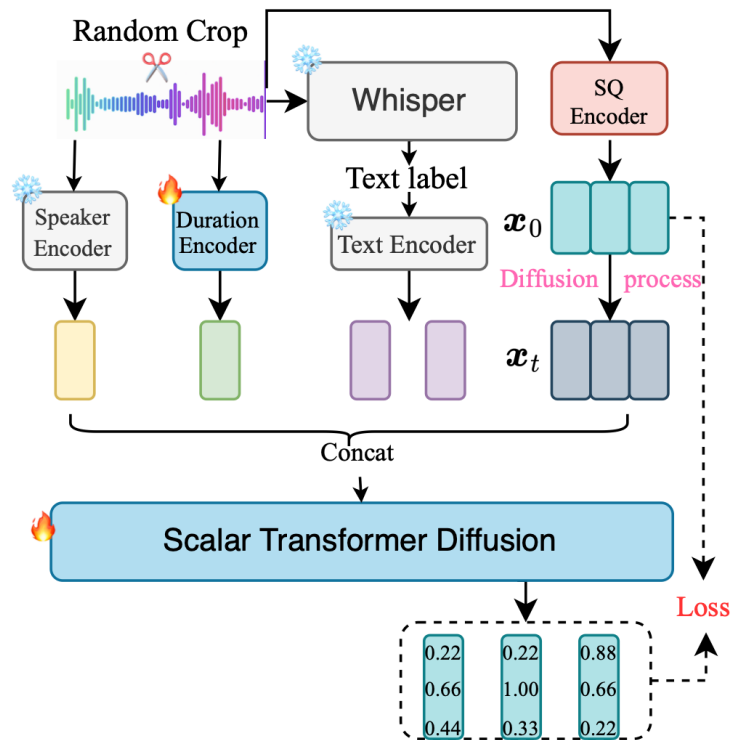
NaturalSpeech 3

	Training Data	Sim-O $\uparrow$	Sim-R $\uparrow$	WER $\downarrow$	CMOS $\uparrow$	SMOS $\uparrow$
Ground Truth	-	0.68	-	1.94	+0.08	3.85
VALL-E $\heartsuit$	Librilight	-	0.58	5.90	-	-
VALL-E $\diamondsuit$	Librilight	0.47	0.51	6.11	-0.60	3.46
NaturalSpeech 2 $\spadesuit$	Librilight	0.55	0.62	1.94	-0.18	3.65
Voicebox $\spadesuit$	Self-Collected (60kh)	0.64	0.67	2.03	-0.23	3.69
Voicebox $\diamondsuit$	Librilight	0.48	0.50	2.14	-0.32	3.52
Mega-TTS 2 $\spadesuit$	Librilight	0.53	-	2.32	-0.20	3.63
UniAudio $\spadesuit$	Mixed (165kh)	0.57	0.68	2.49	-0.25	3.71
StyleTTS 2 $\clubsuit$	LT + V + LJ	0.38	-	2.49	-0.21	3.07
HierSpeech++ $\clubsuit$	LT + LL [*] + EX + MS + NI	0.51	-	6.33	-0.41	3.50
NaturalSpeech 3	Librilight	0.67	0.76	1.81	0.00	4.01

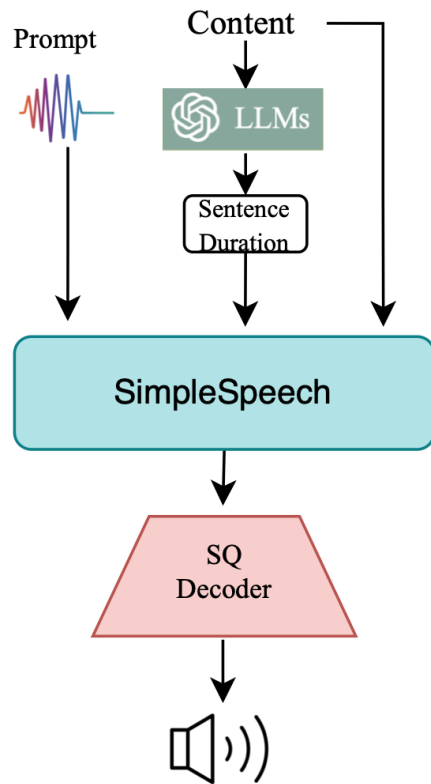
SimpleSpeech



SimpleSpeech



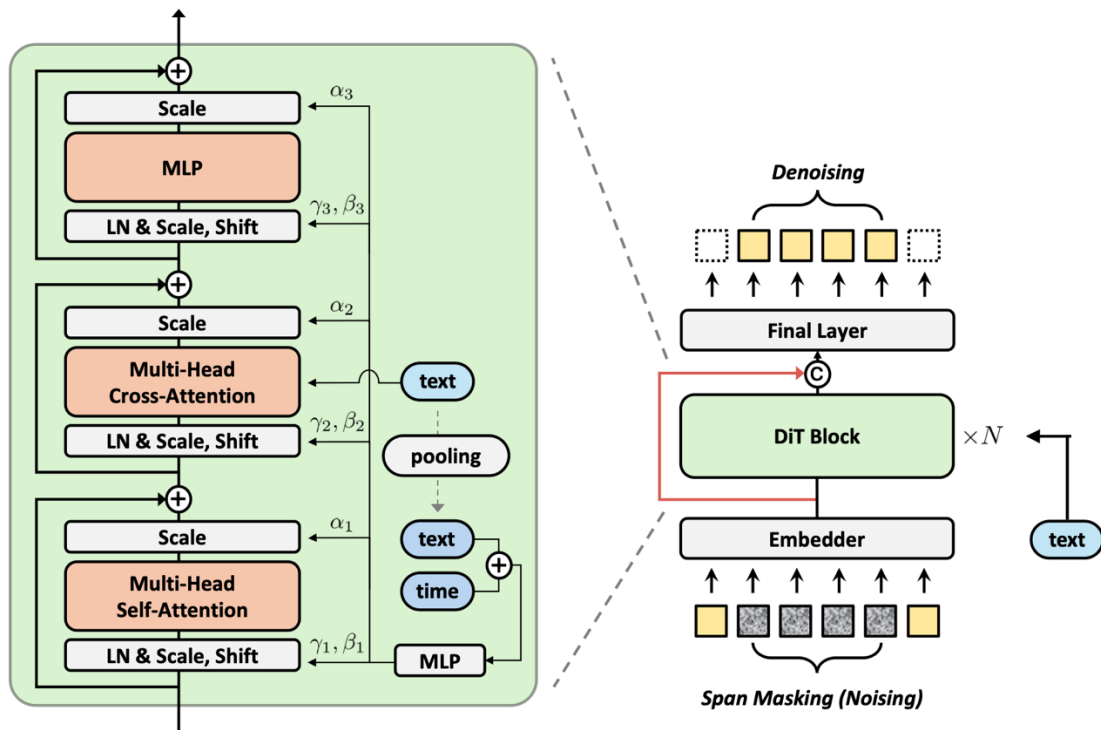
SimpleSpeech



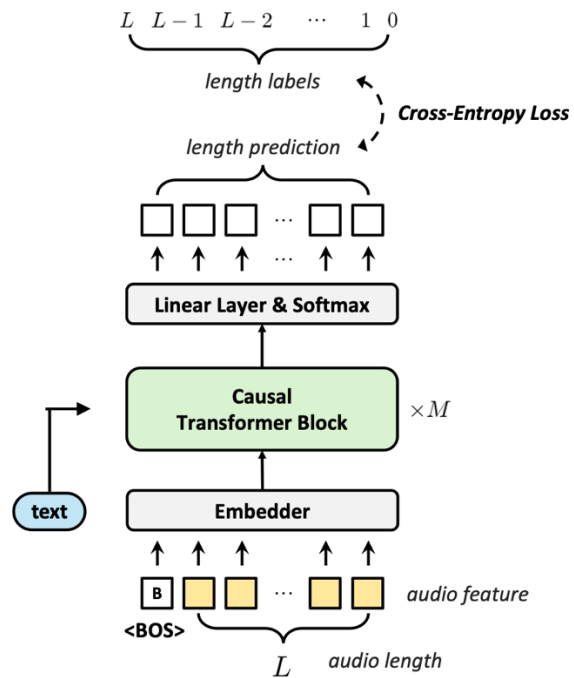
SimpleSpeech

Model	E3TTS	NaturalSpeech 2	SimpleSpeech
MOS	4.32 ± 0.08	4.26 ± 0.08	4.38 ± 0.10
SMOS	4.38 ± 0.17	4.38 ± 0.14	4.30 ± 0.11

DITTO-TTS



DITTO-TTS



DITTO-TTS

Model	SMOS	CMOS
Voicebox (Le et al., 2023) DiTTo-en	2.87 ± 0.45 3.57 ± 0.39	0.14 0.0
VALL-E (Wang et al., 2023) DiTTo-en	3.50 ± 0.46 3.90 ± 0.38	-0.94 0.0
MegaTTS (Jiang et al., 2023) DiTTo-en	3.76 ± 0.44 3.82 ± 0.32	-0.31 0.0
E3-TTS (Gao et al., 2023) DiTTo-en	4.25 ± 0.19 4.28 ± 0.32	-0.30 0.0
NaturalSpeech 2 (Shen et al., 2024) DiTTo-en	3.99 ± 0.37 4.01 ± 0.35	-0.29 0.0
NaturalSpeech 3 (Ju et al., 2024) DiTTo-en	4.42 ± 0.28 3.92 ± 0.27	-0.27 0.0

Diffusion Text-to-Music

- AudioLDM
- Stable Audio
- DiffRhythm
- ACE-Step

Important papers

- [Generative Modeling by Estimating Gradients of the Data Distribution](#)
- [Denoising Diffusion Probabilistic Models](#)
- [Score-Based Generative Modeling through Stochastic Differential Equations](#)
- [WaveGrad: Estimating Gradients for Waveform Generation](#)
- [DiffWave: A Versatile Diffusion Model for Audio Synthesis](#)
- [Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech](#)

Additional papers

- [Diff-TTS: A Denoising Diffusion Model for Text-to-Speech](#)
- [ProDiff: Progressive Fast Diffusion Model For High-Quality Text-to-Speech](#)
- [DiffGAN-TTS: High-Fidelity and Efficient Text-to-Speech with Denoising Diffusion GANs](#)
- [BDDM: Bilateral Denoising Diffusion Models for Fast and High-Quality Speech Synthesis](#)
- [SpecGrad: Diffusion Probabilistic Model based Neural Vocoder with Adaptive Noise Spectral Shaping](#)
- [WaveGrad 2: Iterative Refinement for Text-to-Speech Synthesis](#)

Contacts



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